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1 Reminder

1.1 Observations and Tricks

- Contribution Technique
- 二分圖/Spanning Tree/DFS Tree
- 行、列操作互相獨立
- 奇偶性
- 當 s, t 遞增並且 $t = f(s)$ ，對 s 二分搜不好做，可以改成對 t 二分搜，再算 $f(t)$
- 啟發式合併
- Permutation Normalization (做一些平移對齊兩個 permutation)
- 枚舉 $a_1 \sim a_n$ 再枚舉 $a_n \sim a_1$ 可以包在一個迴圈
- 兩個凸型函數相加還是凸型函數，相減不一定
- 一個區間的 $\text{mex} = k$ ，表示這個區間包含 $0 \sim k-1$ 所有數字，並且「 U - 區間」的最小值 $= k$ 。

1.2 Bug List

- 沒開 long long
- 陣列截出界／陣列開不夠大
- 寫好的函式忘記呼叫
- 0-base / 1-base
- 忘記初始化
- == 打成 =
- <= 打成 <+
- dp[i] 從 dp[i-1] 轉移時忘記特判 $i > 0$
- std::sort 比較運算子寫成 < 或是讓 = 的情況為 true
- 漏 case
- 線段樹改值懶標初始值不能設為 0
- DFS 的時候不小心覆寫到全域變數
- 浮點數誤差
- unsigned int128
- 多筆測資不能沒讀完直接 return
- 記得刪 cerr
- vector 超級肥，小 vector 請用 array，例如矩陣快速冪

2 Init (Linux)

開場流程：

```
1 vim ~/.vimrc
2 mkdir contest && cd contest
3
4 vim template.cpp
5 for c in {A..P}; do
6     cp template.cpp $c.cpp
7 done
8
9 vim run.sh && chmod 777 run.sh
```

2.1 vimrc

```
1 syn on
2 set nu rnu ru cul mouse=a
3 set cin et ts=4 sw=4 sts=4
4 set autochdir
5 set clipboard=unnamedplus
6
7 colo koehler
8
9 no <C-h> ^
10 no <C-l> $
11 no ; :
12
13 inoremap {<CR> {<CR>}<Esc>ko
```

2.2 run.sh

```
1 #!/bin/bash
2
3 g++ -std=c++17 -O2 -g -fsanitize=undefined,address
4 $1 && echo DONE COMPILE || exit 1
5 ./a.out
```

3 Basic

3.1 PBDS

```
1 #include <bits/extc++.h>
2 using namespace __gnu_pbds;
3
4 // map
5 tree<int, int, less<>, rb_tree_tag,
6 tree_order_statistics_node_update> tr;
7 tr.order_of_key(element);
8 tr.find_by_order(rank);
9
10 // set
11 tree<int, null_type, less<>, rb_tree_tag,
12 tree_order_statistics_node_update> tr;
13 tr.order_of_key(element);
14 tr.find_by_order(rank);
15
16 // priority queue
17 __gnu_pbds::priority_queue<int, less<int> >
18 big_q; // Big First
19 __gnu_pbds::priority_queue<int, greater<int> >
20 small_q; // Small First
21 q1.join(q2); // join
```

3.2 Random

```
1 mt19937
2 gen(chrono::steady_clock::now().time_since_epoch().count());
3 #define RANDINT(a, b) uniform_int_distribution<int>
4 (a, b)(rng) // inclusive
5 #define RANDLL(a, b) uniform_int_distribution<long
6 long>(a, b)(rng) // inclusive
```

```
4 #define RANDFLOAT(a, b)
5 uniform_real_distribution<float>(a, b)(rng) //
6 exclusive
7 #define RANDDOUBLE(a, b)
8 uniform_real_distribution<double>(a, b)(rng) //
9 exclusive
10 shuffle(v.begin(), v.end(), gen);
```

4 Python

4.1 I/O

```
1 import sys
2 input = sys.stdin.readline
3
4 # Input
5 def readInt():
6     return int(input())
7 def readIntList():
8     return list(map(int, input().split()))
9 def readStr():
10    s = input()
11    return list(s[:len(s) - 1])
12 def readInts():
13    return map(int, input().split())
```

4.2 Decimal

```
1 from decimal import *
2 getcontext().prec = 500 # precision
3 getcontext().Emax = 500 # 科學記號指數最大值
4 # 將東西轉成 Decimal
5 Decimal(x)
6 Decimal(y)
7 Decimal(0.0)
8 # 運算子跟一般浮點數一樣
9 x *= y
10 # 輸出
11 print(x.quantize(Decimal("0.000001"),
12 rounding=ROUND_HALF_EVEN))
13 # ROUND_CEILING (2.9=>3, -2.1=>-2)
14 # ROUND_FLOOR (2.1=>2, -2.9=>-3)
15 # ROUND_HALF_EVEN (2.5=>2, 3.5=>4)
16 # ROUND_HALF_UP (2.5=>3, -2.5=>-3)
17 # ROUND_HALF_DOWN (2.5=>2, -2.5=>-2)
18 # ROUND_UP (2.1=>3, -2.1=>-3)
19 # ROUND_DOWN (2.9=>2, -2.9=>-2)
```

5 Data Structure

5.1 CDQ

```
1 // preparation
2 // 1. write a 1-base BIT
3 // 2. init(n, mxc): n is number of elements, mxc =
4 值域
5 // 3. build info array CDQ.v by yourself (x, y, z,
6 id)
7 // 4. call solve()
8 // => cdq.ans[i] is number of j s.t. (xj <= xi, yj
9 <= yi, zj <= zi) (j != i)
10 struct CDQ {
11     struct info {
12         int x, y, z, id;
13         info(int x, int y, int z, int id):
14             x(x), y(y), z(z), id(id) {}
15         info():
16             x(0), y(0), z(0), id(0) {}
17     };
18     int n, mxc;
```

```

16 BIT bit;
17 vector<info> v;
18 vector<ll> ans;
19 void init(int _n, int _mxc) {
20     n = _n; mxc = _mxc;
21     v.assign(n, info{});
22     ans.assign(n, 0);
23 }
24 void dfs(int l, int r) {
25     if (l >= r) return;
26     int mid = (l+r) >> 1;
27     dfs(l, mid); dfs(mid+1, r);
28
29     vector<info> tmp;
30     vector<int> bit_op;
31     int pl = l, pr = mid+1;
32
33     while (pl <= mid && pr <= r) {
34         if (v[pl].y <= v[pr].y) {
35             tmp.emplace_back(v[pl]);
36             bit.upd(v[pl].z, 1);
37             bit_op.emplace_back(v[pl].z);
38             pl++;
39         }
40         else {
41             tmp.emplace_back(v[pr]);
42             ans[v[pr].id] += bit.qry(v[pr].z);
43             pr++;
44         }
45     }
46     while (pl <= mid) {
47         tmp.emplace_back(v[pl]);
48         pl++;
49     }
50     while (pr <= r) {
51         tmp.emplace_back(v[pr]);
52         ans[v[pr].id] += bit.qry(v[pr].z);
53         pr++;
54     }
55     for (int i = l, j = 0; i <= r; i++, j++)
56         v[i] = tmp[j];
57     for (auto& op : bit_op) bit.upd(op, -1);
58     void solve() {
59         bit.init(mxc);
60         sort(v.begin(), v.end(), [&](const info& a,
61 const info& b){
62             return (a.x == b.x ? (a.y == b.y ? (a.z
63 == b.z ? a.id < b.id : a.z < b.z) : a.y < b.y) :
64 a.x < b.x);
65         });
66         dfs(0, n-1);
67         map<pair<int, pii>, int> s;
68         sort(v.begin(), v.end(), [&](const info& a,
69 const info& b){
70             return a.id > b.id;
71         });
72         for (int i = 0, id = n-1; i < n; i++, id--)
73             {
74                 pair<int, pii> tmp = make_pair(v[i].x,
75 make_pair(v[i].y, v[i].z));
76                 auto it = s.find(tmp);
77                 if (it != s.end())
78                     ans[id] += it->second;
79                 s[tmp]++;
80             }
81     }
82 }

```

```

74     }
75 }
76 };

```

5.2 Mo's Algorithm

```

1 // segments are 0-based
2 ll cur = 0; // current answer
3 int pl = 0, pr = -1;
4 for (auto& qi : Q) {
5     // get (l, r, qid) from qi
6     while (pl < l) del(pl++);
7     while (pl > l) add(--pl);
8     while (pr < r) add(++pr);
9     while (pr > r) del(pr--);
10    ans[qid] = cur;
11 }

```

5.3 Heavy Light Decomposition

```

1 // Author: Ian
2 void build(V<int>&v);
3 void modify(int p, int k);
4 int query(int ql, int qr);
5 // Insert [ql, qr] segment tree here
6 inline void solve(){
7     int n, q; cin >> n >> q;
8     V<int> v(n);
9     for (auto& i: v) cin >> i;
10    V<V<int>> e(n);
11    for(int i = 1; i < n; i++){
12        int a, b; cin >> a >> b, a--, b--;
13        e[a].emplace_back(b);
14        e[b].emplace_back(a);
15    }
16    V<int> d(n, 0), f(n, 0), sz(n, 1), son(n, -1);
17    F<void(int, int)> dfs1 = [&](int x, int pre) {
18        for (auto i: e[x]) if (i != pre) {
19            d[i] = d[x]+1, f[i] = x;
20            dfs1(i, x), sz[x] += sz[i];
21            if (son[x] == -1 || sz[son[x]] < sz[i])
22                son[x] = i;
23        }
24    }; dfs1(0,0);
25    V<int> top(n, 0), dfn(n, -1);
26    F<void(int,int)> dfs2 = [&](int x, int t) {
27        static int cnt = 0;
28        dfn[x] = cnt++, top[x] = t;
29        if (son[x] == -1) return;
30        dfs2(son[x], t);
31        for (auto i: e[x]) if (!~dfn[i])
32            dfs2(i, i);
33    }; dfs2(0,0);
34    V<int> dfnv(n);
35    for (int i = 0; i < n; i++)
36        dfnv[dfn[i]] = v[i];
37    build(dfnv);
38    while(q--){
39        int op, a, b, ans; cin >> op >> a >> b;
40        switch(op){
41            case 1:
42                modify(dfn[a-1], b);
43                break;
44            case 2:
45                a--, b--, ans = 0;
46                while (top[a] != top[b]) {

```

```

47     if (d[top[a]] > d[top[b]]) swap(a,b);
48     ans = max(ans, query(dfn[top[b]],
49         dfn[b]+1));
50     b = f[top[b]];
51 }
52 if (dfn[a] > dfn[b]) swap(a,b);
53 ans = max(ans, query(dfn[a], dfn[b]+1));
54 cout << ans << endl;
55 break;
56 }
57 }

```

5.4 Leftist Heap

```

1 // Author: Ian
2 // Function: min-heap, with worst-time  $O(\lg n)$ 
3 merge
4 struct node {
5     node *l, *r; int d, v;
6     node(int x): d(1), v(x) { l = r = nullptr; }
7 };
8 static inline int d(node* x) { return x ? x->d : 0; }
9 node* merge(node* a, node* b) {
10     if (!a || !b) return a ? a : b;
11     if (a->v > b->v) swap(a,b);
12     a->r = merge(a->r, b);
13     if (d(a->l) < d(a->r))
14         swap(a->l, a->r);
15     a->d = d(a->r) + 1;
16     return a;
17 }

```

5.5 Persistent Treap

```

1 // Author: Ian
2 struct node {
3     node *l, *r;
4     char c; int v, sz;
5     node(char x = '$'): c(x), v(mt()), sz(1) {
6         l = r = nullptr;
7     }
8     node(node* p) { *this = *p; }
9     void pull() {
10         sz = 1;
11         for (auto i : {l, r})
12             if (i) sz += i->sz;
13     }
14 } arr[maxn], *ptr = arr;
15 inline int size(node* p) { return p ? p->sz : 0; }
16 node* merge(node* a, node* b) {
17     if (!a || !b) return a ? a : b;
18     if (a->v < b->v) {
19         node* ret = new(ptr++) node(a);
20         ret->r = merge(ret->r, b); ret->pull();
21         return ret;
22     }
23     else {
24         node* ret = new(ptr++) node(b);
25         ret->l = merge(a, ret->l); ret->pull();
26         return ret;
27     }
28 }
29 P<node*> split(node* p, int k) {
30     if (!p) return {nullptr, nullptr};
31     if (k >= size(p->l) + 1) {

```

```

32     auto [a, b] = split(p->r, k - size(p->l) - 1);
33     node* ret = new(ptr++) node(p);
34     ret->r = a, ret->pull();
35     return {ret, b};
36 }
37 else {
38     auto [a, b] = split(p->l, k);
39     node* ret = new(ptr++) node(p);
40     ret->l = b, ret->pull();
41     return {a, ret};
42 }
43 }

```

5.6 Li Chao Tree

```

1 // Author: Ian
2 // Function: For a set of lines L, find the maximum
3 L_i(x) in L in  $O(\lg n)$ .
4 typedef long double ld;
5 constexpr int maxn = 5e4 + 5;
6 struct line {
7     ld a, b;
8     ld operator()(ld x) { return a * x + b; }
9 } arr[(maxn + 1) << 2];
10 bool operator<(line a, line b) { return a.a < b.a; }
11 #define m ((l+r)>>1)
12 void insert(line x, int i = 1, int l = 0, int r = maxn) {
13     if (r - l == 1) {
14         if (x(l) > arr[i](l))
15             arr[i] = x;
16         return;
17     }
18     line a = max(arr[i], x), b = min(arr[i], x);
19     if (a(m) > b(m))
20         arr[i] = a, insert(b, i << 1, l, m);
21     else
22         arr[i] = b, insert(a, i << 1 | 1, m, r);
23 }
24 ld query(int x, int i = 1, int l = 0, int r = maxn) {
25     if (x < l || r <= x) return -
26         numeric_limits<ld>::max();
27     if (r - l == 1) return arr[i](x);
28     return max({arr[i](x), query(x, i << 1, l, m),
29         query(x, i << 1 | 1, m, r)});
30 }
31 #undef m

```

5.7 Time Segment Tree

```

1 // Author: Ian
2 constexpr int maxn = 1e5 + 5;
3 V<P<int>> arr[(maxn + 1) << 2];
4 V<int> dsu, sz;
5 V<tuple<int, int, int>> his;
6 int cnt, q;
7 int find(int x) {
8     return x == dsu[x] ? x : find(dsu[x]);
9 }
10 inline bool merge(int x, int y) {
11     int a = find(x), b = find(y);
12     if (a == b) return false;
13     if (sz[a] > sz[b]) swap(a, b);
14     his.emplace_back(a, b, sz[b]), dsu[a] = b,
15     sz[b] += sz[a];
16     return true;
17 }

```

```

17 inline void undo() {
18     auto [a, b, s] = his.back(); his.pop_back();
19     dsu[a] = a, sz[b] = s;
20 }
21 #define m ((l + r) >> 1)
22 void insert(int ql, int qr, P<int> x, int i = 1,
23             int l = 0, int r = q) {
24     // debug(ql, qr, x); return;
25     if (qr <= l || r <= ql) return;
26     if (ql <= l && r <= qr) {arr[i].push_back(x);
27         return;}
28     if (qr <= m)
29         insert(ql, qr, x, i << 1, l, m);
30     else if (m <= ql)
31         insert(ql, qr, x, i << 1 | 1, m, r);
32     else {
33         insert(ql, qr, x, i << 1, l, m);
34         insert(ql, qr, x, i << 1 | 1, m, r);
35     }
36 }
37 void traversal(V<int>& ans, int i = 1, int l = 0,
38              int r = q) {
39     int opcnt = 0;
40     // debug(i, l, r);
41     for (auto [a, b] : arr[i])
42         if (merge(a, b))
43             opcnt++, cnt--;
44     if (r - l == 1) ans[l] = cnt;
45     else {
46         traversal(ans, i << 1, l, m);
47         traversal(ans, i << 1 | 1, m, r);
48     }
49     while (opcnt--)
50         undo(), cnt++;
51     arr[i].clear();
52 }
53 #undef m
54 inline void solve() {
55     int n, m; cin >> n >> m >> q, q++;
56     dsu.resize(cnt = n), sz.assign(n, 1);
57     iota(dsu.begin(), dsu.end(), 0);
58     // a, b, time, operation
59     unordered_map<ll, V<int>> s;
60     for (int i = 0; i < m; i++) {
61         int a, b; cin >> a >> b;
62         if (a > b) swap(a, b);
63         s[((ll)a << 32) | b].emplace_back(0);
64     }
65     for (int i = 1; i < q; i++) {
66         int op, a, b;
67         cin >> op >> a >> b;
68         if (a > b) swap(a, b);
69         switch (op) {
70             case 1:
71                 s[((ll)a << 32) | b].push_back(i);
72                 break;
73             case 2:
74                 auto tmp = s[((ll)a << 32) | b].back();
75                 s[((ll)a << 32) | b].pop_back();
76                 insert(tmp, i, P<int> {a, b});
77         }
78     }
79     for (auto [p, v] : s) {
80         int a = p >> 32, b = p & -1;

```

```

78         while (v.size()) {
79             insert(v.back(), q, P<int> {a, b});
80             v.pop_back();
81         }
82     }
83     V<int> ans(q);
84     traversal(ans);
85     for (auto i : ans)
86         cout << i << ' ';
87     cout << endl;
88 }

```

6 DP

- 區間 DP
 - 狀態： $dp[l][r]$ = 區間 $[l, r]$ 的最佳值/方案數
 - 轉移：枚舉劃分點 k
 - 思考：是否滿足四邊形不等式、Knuth 優化可加速
- 背包 DP
 - 狀態： $dp[i][w]$ = 前 i 個物品容量 w 的最佳值
 - 判斷是 0/1、多重、分組 $\Rightarrow$ 決定轉移方式
 - 若容量大 $\Rightarrow$ bitset / 數學變形 / meet-in-the-middle
- 樹形 DP
 - 狀態： $dp[u][flag]$ = 子樹 u 的最佳值
 - 合併子樹資訊 $\Rightarrow$ 小到大合併 / 捲積式轉移
 - 注意 reroot 技巧 (dp on tree + dp2 上傳)
- 數位 DP
 - 狀態： $(pos, tight, property)$
 - tight 控制是否貼上界
 - property 常為「餘數、數字和、相鄰限制」
- 狀態 DP
 - 狀態： $dp[mask][last]$
 - 常見於 TSP / Hamiltonian path / 覆蓋問題
 - $n \leq 20$ 可做，否則要容斥 / FFT
- 期望 / 機率 DP
 - 狀態 $E[s]$ = 從狀態 s 到終點的期望
 - 式子： $E[s] = c + \sum P(s \rightarrow s')E[s']$
 - 線性期望：能拆就拆，少算分布
 - 輸出 mod $\Rightarrow$ 分數化 $\Rightarrow$ 模逆元
- 計數 DP / 組合數
 - 狀態表示方案數，常搭配「模數取餘」
 - 若轉移是捲積型 $\Rightarrow$ FFT/NTT 加速
 - 若能公式化 (Catalan / Ballot / Stirling) $\Rightarrow$ 直接套公式
- 優化 DP
 - 判斷轉移方程 $dp[i] = \min_j (dp[j] + C(j, i))$ 的性質
 - 單調性 $\Rightarrow$ 分治優化
 - 凸性 $\Rightarrow$ Convex Hull Trick / 斜率優化
 - 四邊形不等式 $\Rightarrow$ Knuth 優化

6.1 Aliens

```

1 // Author: Gino
2 // Function: TODO
3 int n; ll k;
4 vector<ll> a;
5 vector<pll> dp[2];
6 void init() {
7     cin >> n >> k;
8     for (auto& d : dp) d.clear(), d.resize(n);
9     a.clear(); a.resize(n);
10    for (auto& i : a) cin >> i;
11 }
12 pll calc(ll p) {
13     dp[0][0] = make_pair(0, 0);

```

```

14 dp[1][0] = make_pair(-a[0], 0);
15 for (int i = 1; i < n; i++) {
16     if (dp[0][i-1].first > dp[1][i-1].first + a[i]
17         - p) {
18         dp[0][i] = dp[0][i-1];
19     } else if (dp[0][i-1].first < dp[1][i-1].first
20         + a[i] - p) {
21         dp[0][i] = make_pair(dp[1][i-1].first + a[i]
22             - p, dp[1][i-1].second+1);
23     } else {
24         dp[0][i] = make_pair(dp[0][i-1].first,
25             min(dp[0][i-1].second, dp[1][i-1].second+1));
26     }
27     if (dp[0][i-1].first - a[i] > dp[1][i-1].first)
28     {
29         dp[1][i] = make_pair(dp[0][i-1].first - a[i],
30             dp[0][i-1].second);
31     } else if (dp[0][i-1].first - a[i] < dp[1][i-1].first)
32     {
33         dp[1][i] = dp[1][i-1];
34     } else {
35         dp[1][i] = make_pair(dp[1][i-1].first,
36             min(dp[0][i-1].second, dp[1][i-1].second));
37     }
38 }
39 return dp[0][n-1];
40 }
41 void solve() {
42     ll l = 0, r = 1e7;
43     pll res = calc(0);
44     if (res.second <= k) return cout << res.first << endl, void();
45     while (l < r) {
46         ll mid = (l+r)>>1;
47         res = calc(mid);
48         if (res.second <= k) r = mid;
49         else l = mid+1;
50     }
51     res = calc(l);
52     cout << res.first + k*l << endl;
53 }

```

6.2 SOS DP

```

1 // Author: Gino
2 // Function: Solve problems that enumerates subsets
3 // of subsets (3^n => n*2^n)
4 for (int msk = 0; msk < (1<<n); msk++) {
5     for (int i = 1; i <= n; i++) {
6         if (msk & (1<<(i-1))) {
7             // dp[msk][i] = dp[msk][i-1] + dp[msk
8             ^ (1<<(i-1))][i-1];
9         } else {
10             // dp[msk][i] = dp[msk][i-1];
11         }
12     }
13 }

```

6.3 期望 DP (Expected Value DP)

- 狀態設計： $E[s]$ = 從狀態 s 出發到終點的期望值
- 列式子： $E[s] = (\text{當前代價}) + \sum_{s'} P(s \rightarrow s') \cdot E[s']$
- 若存在自環，把 $E[s]$ 移到左邊，整理成 $(1 - P(s \rightarrow s))E[s] = c + \sum_{s' \neq s} P(s \rightarrow s') \cdot E[s']$
- 線性期望技巧：能拆就拆，避免處理整個分布
- 輸出 mod 時，分母要用模逆元： $q^{-1} \equiv q^{M-2} \pmod{M}$ (質數模數)

常見題型

- 擲骰子遊戲 (到達終點的期望步數)
- 隨機遊走 hitting time
- 重複試驗直到成功
- 博弈遊戲的期望值
- 機率 DP：計算到某步時在某狀態的機率

範例：擲骰子到 n 格

$$E[i] = 1 + \frac{1}{6} \sum_{d=1}^6 E[i+d], \quad (i < n), \quad E[n] = 0$$

```

1 int main(){
2     int n;
3     cin >> n; // 終點位置
4
5     // E[i] = 從位置 i 走到終點的期望步數
6     // 因為每次最多走 6，所以要開 n+6 以避免越界
7     vector<double> E(n+7, 0.0);
8
9     // 從終點往前推 (backward DP)
10    for(int i=n-1; i>=0; i--){
11        double sum=0;
12        // 期望公式: E[i] = 1 + (E[i+1]+...
13        +E[i+6]) / 6
14        for(int d=1; d<=6; d++) sum += E[i+d];
15        E[i] = 1 + sum/6.0;
16    }
17
18    // 輸出 E[0]，即從起點到終點的期望擲骰次數
19    cout << fixed << setprecision(10) << E[0] <<
20    "\n";
21 }

```

6.4 數位 DP (Digit DP)

- 狀態： $(pos, tight, property)$
 - pos = 當前處理到第幾位
 - $tight$ = 是否受限於上界 N
 - $property$ = 額外屬性 (如數位和、餘數、相鄰限制...)
- 遞迴：枚舉當前位數字，遞迴下一位
- 終止條件： $pos == \text{長度} \Rightarrow$ 回傳屬性是否滿足
- 記憶化： $dp[pos][tight][property]$

常見題型

- 計算 $[0, N]$ 中數位和可被 k 整除的數字個數
- 不含連續相同數字的數字個數
- 含特定數字次數的數字個數
- 位數和 / 餘數 / mod pattern

範例：計算 $[0, N]$ 中數位和 $\text{mod } k = 0$ 的數字個數

$$dp[pos][tight][\text{sum mod } k]$$

```

1 string s; // N 轉成字串，方便逐位處理
2 int k; // 除數
3
4 // dp[pos][tight][sum_mod]
5 // pos = 當前處理到哪一位 (0 = 最高位)
6 // tight = 是否仍受限於 N 的數字 (1 = 是, 0 = 否)
7 // sum_mod = 當前數位和 mod k 的值
8 long long dp[20][2][105];
9
10 // 計算：從 pos 開始，tight 狀態下，數位和 mod k =
11 // sum_mod 的方案數
12 long long dfs(int pos, int tight, int sum_mod){
13     // 終止條件：所有位數都處理完
14 }

```



```

13     if(pos == (int)s.size())
14         // 若數位和 mod k == 0, 算作一個合法數字
15         return (sum_mod % k == 0);
16
17     // 記憶化查詢
18     if(dp[pos][tight][sum_mod] != -1)
19         return dp[pos][tight][sum_mod];
20
21     long long res = 0;
22     // 如果 tight = 1, 本位數字上限 = N 的該位數字
23     // 如果 tight = 0, 本位數字上限 = 9
24     int limit = tight ? (s[pos] - '0') : 9;
25
26     // 枚舉當前位可以填的數字
27     for(int d=0; d<=limit; d++){
28         // 下一位是否仍然 tight?
29         int next_tight = (tight && d==limit);
30         // 更新數位和 mod k
31         int next_mod = (sum_mod + d) % k;
32         res += dfs(pos+1, next_tight, next_mod);
33     }
34
35     // 存結果
36     return dp[pos][tight][sum_mod] = res;
37 }
38
39 int main(){
40     long long N;
41     cin >> N >> k;
42     s = to_string(N); // 把 N 轉成字串, 方便取每一位
43     memset(dp, -1, sizeof(dp));
44     cout << dfs(0, 1, 0) << "\n"; // 從最高位開始, 初始
45     tight=1, sum=0

```

```

25 }
26 bool Bfs(auto &&g, int r) {
27     iota(all(fa), 0); ranges::fill(s, -1);
28     q = queue<int>(); q.push(r); s[r] = 0;
29     for (; !q.empty(); q.pop()) {
30         for (int x = q.front(); int u : g[x])
31             if (s[u] == -1) {
32                 if (pre[u] = x, s[u] = 1, match[u] == n)
33                     for (int a = u, b = x, last;
34                         b != n; a = last, b = pre[a])
35                         last = match[b], match[b] = a,
36                         match[a] = b;
37                 return true;
38             }
39             q.push(match[u]); s[match[u]] = 0;
40         } else if (!s[u] && Find(u) != Find(x)) {
41             int l = LCA(u, x);
42             Blossom(x, u, l); Blossom(u, x, l);
43         }
44     }
45     return false;
46 }
47 Matching(auto &&g) : ans(0), n((int)g.size()),
48     fa(n+1), s(n+1), v(n+1), pre(n+1, n), match(n+1,
49     n) {
50     for (int x = 0; x < n; ++x)
51         if (match[x] == n) ans += Bfs(g, x);
52 } // match[x] == n means not matched
53 }; // test @ yosupo judge

```

7.2 Weighted Blossom

```

1 struct WeightGraph {
2     static const int INF=INT_MAX, N=514;
3     struct edge{
4         int u, v, w; edge(){}
5         edge(int ui, int vi, int wi):u(ui), v(vi), w(wi){}
6     };
7     int n, n_x, lab[N*2], match[N*2], slack[N*2], st[N*2];
8     int pa[N*2], flo_from[N*2][N+1], S[N*2], vis[N*2];
9     edge g[N*2][N*2]; vector<int> flo[N*2];
10    queue<int> q;
11    int e_delta(const edge &e){
12        return lab[e.u]+lab[e.v]-g[e.u][e.v].w*2;
13    }
14    void update_slack(int u, int x){
15        if(!slack[x] || e_delta(g[u][x]) <
16            e_delta(g[slack[x]][x])) slack[x]=u;
17    }
18    void set_slack(int x){
19        slack[x]=0;
20        for(int u=1; u<=n; ++u)
21            if(g[u][x].w>0 && st[u]!=x && S[st[u]]==0)
22                update_slack(u, x);
23    }
24    void q_push(int x){
25        if(x<=n) q.push(x);
26        else for(size_t i=0; i<flo[x].size(); ++i)
27            q_push(flo[x][i]);
28    }
29    void set_st(int x, int b){
30        st[x]=b;
31        if(x>n) for(size_t i=0; i<flo[x].size(); ++i)
32            set_st(flo[x][i], b);
33    }

```

7 Graph

7.1 Blossom

```

1 // Author: ckiseki
2 // usage: first build 0-based graph
3 // vector<vector<int>> g on your own
4 // then construct the object: Matching M(g)
5 // => M.ans
6 // to check matching status: M.match[x] (== n if
7 // not matched)
8 struct Matching {
9     queue<int> q; int ans, n;
10    vector<int> fa, s, v, pre, match;
11    int Find(int u) {
12        return u == fa[u] ? u : fa[u] = Find(fa[u]); }
13    int LCA(int x, int y) {
14        static int tk = 0; tk++; x = Find(x); y =
15        Find(y);
16        for (; swap(x, y) if (x != n) {
17            if (v[x] == tk) return x;
18            v[x] = tk;
19            x = Find(pre[match[x]]);
20        }
21    }
22    void Blossom(int x, int y, int l) {
23        for (; Find(x) != l; x = pre[y]) {
24            pre[x] = y, y = match[x];
25            if (s[y] == 1) q.push(y), s[y] = 0;
26            for (int z: {x, y}) if (fa[z] == z) fa[z] =
27            l;
28        }
29    }

```

```

33 int get_pr(int b,int xr){
34     int pr=find(flo[b].begin(),flo[b].end(),xr)-
        flo[b].begin();
35     if(pr%2==1){
36         reverse(flo[b].begin()+1,flo[b].end());
37         return (int)flo[b].size()-pr;
38     }else return pr;
39 }
40 void set_match(int u,int v){
41     match[u]=g[u][v].v; if(u<=n) return; edge
        e=g[u][v];
42     int xr=flo_from[u][e.u],pr=get_pr(u,xr);
43     for(int i=0;i<pr;++i)
44         set_match(flo[u][i],flo[u][i^1]);
45     set_match(xr,v);
46     rotate(flo[u].begin(),flo[u].begin()+pr,flo[u].end());
47 }
48 void augment(int u,int v){
49     for(;;){
50         int xnv=st[match[u]]; set_match(u,v);
51         if(!xnv) return;
52         set_match(xnv,st[pa[xnv]]);
53         u=st[pa[xnv]],v=xnv;
54     }
55 }
56 int get_lca(int u,int v){
57     static int t=0;
58     for(++t;u||v;swap(u,v)){
59         if(u==0) continue; if(vis[u]==t) return u;
60         vis[u]=t; u=st[match[u]]; if(u) u=st[pa[u]];
61     }
62     return 0;
63 }
64 void add_blossom(int u,int lca,int v){
65     int b=n+1; while(b<=n_x&&st[b])++b; if(b>n_x)+
        +n_x;
66     lab[b]=0,S[b]=0; match[b]=match[lca];
67     flo[b].clear(); flo[b].push_back(lca);
68     for(int x=u,y;x!=lca;x=st[pa[y]])
69         flo[b].push_back(x),
70         flo[b].push_back(y=st[match[x]]),q_push(y);
71     reverse(flo[b].begin()+1,flo[b].end());
72     for(int x=v,y;x!=lca;x=st[pa[y]])
73         flo[b].push_back(x),
74         flo[b].push_back(y=st[match[x]]),q_push(y);
75     set_st(b,b);
76     for(int x=1;x<=n_x;++x) g[b][x].w=g[x][b].w=0;
77     for(int x=1;x<=n;++x) flo_from[b][x]=0;
78     for(size_t i=0;i<flo[b].size();++i){
79         int xs=flo[b][i];
80         for(int x=1;x<=n_x;++x)
81             if(g[b][x].w==0||e_delta(g[xs][x])<e_delta(
82                 g[b][x])) g[b][x]=g[xs][x],g[x][b]=g[x]
83                 [xs];
84         for(int x=1;x<=n;++x)
85             if(flo_from[xs][x])flo_from[b][x]=xs;
86     }
87     set_slack(b);
88 }
89 void expand_blossom(int b){
90     for(size_t i=0;i<flo[b].size();++i)
91         set_st(flo[b][i],flo[b][i]);
92     int xr=flo_from[b][g[b]
93     [pa[b]].u],pr=get_pr(b,xr);
94     for(int i=0;i<pr;i+=2){
95         int xs=flo[b][i],xns=flo[b][i+1];
96         pa[xs]=g[xns][xs].u; S[xs]=1,S[xns]=0;
97         slack[xs]=0,set_slack(xns); q_push(xns);
98     }
99     S[xr]=1,pa[xr]=pa[b];
100     for(size_t i=pr+1;i<flo[b].size();++i){
101         int xs=flo[b][i]; S[xs]=-1,set_slack(xs);
102     }
103     st[b]=0;
104 }
105 bool on_found_edge(const edge &e){
106     int u=st[e.u],v=st[e.v];
107     if(S[v]==-1){
108         pa[v]=e.u,S[v]=1; int nu=st[match[v]];
109         slack[v]=slack[nu]=0; S[nu]=0,q_push(nu);
110     }else if(S[v]==0){
111         int lca=get_lca(u,v);
112         if(!lca) return
113         augment(u,v),augment(v,u),true;
114         else add_blossom(u,lca,v);
115     }
116     return false;
117 }
118 bool matching(){
119     memset(S+1,-1,sizeof(int)*n_x);
120     memset(slack+1,0,sizeof(int)*n_x);
121     q=queue<int>();
122     for(int x=1;x<=n_x;++x)
123         if(st[x]==x&&!
124         match[x])pa[x]=0,S[x]=0,q_push(x);
125     if(q.empty()) return false;
126     for(;;){ while(q.size()){
127         int
128         u=q.front();q.pop();if(S[st[u]]==1)continue;
129         for(int v=1;v<=n;++v)
130             if(g[u][v].w>0&&st[u]!=st[v]){
131                 if(e_delta(g[u][v])==0){
132                     if(on_found_edge(g[u][v])) return
133                     true;
134                 }else update_slack(u,st[v]);
135             }
136     }
137     int d=INF;
138     for(int b=n+1;b<=n_x;++b)
139         if(st[b]==b&&S[b]==1) d=min(d,lab[b]/2);
140     for(int x=1;x<=n_x;++x)
141         if(st[x]==x&&slack[x]){
142             if(S[x]==-1) d=min(d,e_delta(g[slack[x]]
143             [x]));
144             else if(S[x]==0)
145                 d=min(d,e_delta(g[slack[x]][x])/2);
146         }
147     for(int u=1;u<=n;++u){ if(S[st[u]]==0){
148         if(lab[u]<=d) return 0; lab[u]-=d;
149     }else if(S[st[u]]==1) lab[u]+=d;
150 }
151     for(int b=n+1;b<=n_x;++b) if(st[b]==b){
152         if(S[st[b]]==0) lab[b]+=d*2;
153         else if(S[st[b]]==1) lab[b]-=d*2;
154     }
155     q=queue<int>();
156     for(int x=1;x<=n_x;++x)
157         if(st[x]==x&&slack[x]&&st[slack[x]]!
158         =x&&e_delta(g[slack[x]][x])==0)
159             if(on_found_edge(g[slack[x]][x])) return
160             true;

```



```

148     for(int b=n+1;b<=n_x;++b)
149     if(st[b]==b&&S[b]==1&&
150         lab[b]==0) expand_blossom(b);
151     }
152     return false;
153 }
154 pair<long long,int> solve(){
155     memset(match+1,0,sizeof(int)*n); n_x=n;
156     int n_matches=0,w_max=0; long long
157     tot_weight=0;
158     for(int u=0;u<=n;++u) st[u]=u,flo[u].clear();
159     for(int u=1;u<=n;++u) for(int v=1;v<=n;++v){
160         flo_from[u][v]=(u==v?0);
161         w_max=max(w_max,g[u][v].w);
162     }
163     for(int u=1;u<=n;++u) lab[u]=w_max;
164     while(matching()) ++n_matches;
165     for(int u=1;u<=n;++u) if(match[u]&&match[u]<u)
166         tot_weight+=g[u][match[u]].w;
167     return make_pair(tot_weight,n_matches);
168 }
169 void add_edge(int ui,int vi,int wi)
170 { g[ui][vi].w=g[vi][ui].w=wi; }
171 void init(int _n){
172     n=_n;
173     for(int u=1;u<=n;++u) for(int v=1;v<=n;++v)
174         g[u][v]=edge(u,v,0);
175 }
176 }graph;

```

7.3 Max Clique

```

1 // max N = 64
2 typedef unsigned long long ll;
3 struct MaxClique{
4     ll nb[ N ] , n , ans;
5     void init( ll _n ){
6         n = _n;
7         for( int i = 0 ; i < n ; i ++ ) nb[ i ] = 0LLU;
8     }
9     void add_edge( ll _u , ll _v ){
10         nb[ _u ] |= ( 1LLU << _v );
11         nb[ _v ] |= ( 1LLU << _u );
12     }
13     void B( ll r , ll p , ll x , ll cnt , ll res ){
14         if( cnt + res < ans ) return;
15         if( p == 0LLU && x == 0LLU ){
16             if( cnt > ans ) ans = cnt;
17             return;
18         }
19         ll y = p | x; y &= -y;
20         ll q = p & ( ~nb[ int( log2( y ) ) ] );
21         while( q ){
22             ll i = int( log2( q & (-q) ) );
23             B( r | ( 1LLU << i ) , p & nb[ i ] , x &
24                 nb[ i ] , cnt + 1LLU , __builtin_popcountll( p &
25                 nb[ i ] ) );
26             q &= ~( 1LLU << i );
27             p &= ~( 1LLU << i );
28             x |= ( 1LLU << i );
29         }
30     }
31     int solve(){
32         ans = 0;
33         ll _set = 0;

```

```

33         if( n < 64 ) _set = ( 1LLU << n ) - 1;
34         else{
35             for( ll i = 0 ; i < n ; i ++ ) _set |= ( 1LLU
36                 << i );
37             B( 0LLU , _set , 0LLU , 0LLU , n );
38             return ans;
39         }
40     }maxClique;

```

7.4 Bellman-Ford + SPFA

```

1 int n , m;
2
3 // Graph
4 vector<vector<pair<int, ll> > > g;
5 vector<ll> dis;
6 vector<bool> negCycle;
7
8 // SPFA
9 vector<int> rlx;
10 queue<int> q;
11 vector<bool> inq;
12 vector<int> pa;
13 void SPFA(vector<int>& src) {
14     dis.assign(n+1, LINF);
15     negCycle.assign(n+1, false);
16     rlx.assign(n+1, 0);
17     while (!q.empty()) q.pop();
18     inq.assign(n+1, false);
19     pa.assign(n+1, -1);
20
21     for (auto& s : src) {
22         dis[s] = 0;
23         q.push(s); inq[s] = true;
24     }
25
26     while (!q.empty()) {
27         int u = q.front();
28         q.pop(); inq[u] = false;
29         if (rlx[u] >= n) {
30             negCycle[u] = true;
31         }
32         else for (auto& e : g[u]) {
33             int v = e.first;
34             ll w = e.second;
35             if (dis[v] > dis[u] + w) {
36                 dis[v] = dis[u] + w;
37                 rlx[v] = rlx[u] + 1;
38                 pa[v] = u;
39                 if (!inq[v]) {
40                     q.push(v);
41                     inq[v] = true;
42                 }
43             }
44         }
45     }
46
47     // Bellman-Ford
48     queue<int> q;
49     vector<int> pa;
50     void BellmanFord(vector<int>& src) {
51         dis.assign(n+1, LINF);
52         negCycle.assign(n+1, false);
53         pa.assign(n+1, -1);
54
55         for (auto& s : src) dis[s] = 0;

```

```

55     for (int rlx = 1; rlx <= n; rlx++) {
56         for (int u = 1; u <= n; u++) {
57             if (dis[u] == LINF) continue; //
Important!!
58             for (auto& e : g[u]) {
59                 int v = e.first; ll w = e.second;
60                 if (dis[v] > dis[u] + w) {
61                     dis[v] = dis[u] + w;
62                     pa[v] = u;
63                     if (rlx == n) negCycle[v] =
true;
64             } } } } }
65
66
67 // Negative Cycle Detection
68 void NegCycleDetect() {
69     /* No Neg Cycle: NO
70     Exist Any Neg Cycle:
71     YES
72     v0 v1 v2 ... vk v0 */
73
74     vector<int> src;
75     for (int i = 1; i <= n; i++)
76         src.emplace_back(i);
77
78     SPFA(src);
79     // BellmanFord(src);
80
81     int ptr = -1;
82     for (int i = 1; i <= n; i++) if (negCycle[i])
83         { ptr = i; break; }
84
85     if (ptr == -1) { return cout << "NO" << endl,
void(); }
86
87     cout << "YES\n";
88     vector<int> ans;
89     vector<bool> vis(n+1, false);
90
91     while (true) {
92         ans.emplace_back(ptr);
93         if (vis[ptr]) break;
94         vis[ptr] = true;
95         ptr = pa[ptr];
96     }
97     reverse(ans.begin(), ans.end());
98
99     vis.assign(n+1, false);
100    for (auto& x : ans) {
101        cout << x << ' ';
102        if (vis[x]) break;
103        vis[x] = true;
104    }
105    cout << endl;
106 }
107
108 // Distance Calculation
109 void calcDis(int s) {
110     vector<int> src;
111     src.emplace_back(s);
112     SPFA(src);
113     // BellmanFord(src);
114
115     while (!q.empty()) q.pop();
116     for (int i = 1; i <= n; i++)
117         if (negCycle[i]) q.push(i);

```

```

118
119     while (!q.empty()) {
120         int u = q.front(); q.pop();
121         for (auto& e : g[u]) {
122             int v = e.first;
123             if (!negCycle[v]) {
124                 q.push(v);
125                 negCycle[v] = true;
126             } } } }

```

7.5 BCC - AP

```

1 int n, m;
2 int low[maxn], dfn[maxn], instp;
3 vector<int> E, g[maxn];
4 bitset<maxn> isap;
5 bitset<maxm> vis;
6 stack<int> stk;
7 int bccnt;
8 vector<int> bcc[maxn];
9 inline void popout(int u) {
10     bccnt++;
11     bcc[bccnt].emplace_back(u);
12     while (!stk.empty()) {
13         int v = stk.top();
14         if (u == v) break;
15         stk.pop();
16         bcc[bccnt].emplace_back(v);
17     }
18 }
19 void dfs(int u, bool rt = 0) {
20     stk.push(u);
21     low[u] = dfn[u] = ++instp;
22     int kid = 0;
23     Each(e, g[u]) {
24         if (vis[e]) continue;
25         vis[e] = true;
26         int v = E[e]^u;
27         if (!dfn[v]) {
28             // tree edge
29             kid++; dfs(v);
30             low[u] = min(low[u], low[v]);
31             if (!rt && low[v] >= dfn[u]) {
32                 // bcc found: u is ap
33                 isap[u] = true;
34                 popout(u);
35             }
36         } else {
37             // back edge
38             low[u] = min(low[u], dfn[v]);
39         }
40     }
41     // special case: root
42     if (rt) {
43         if (kid > 1) isap[u] = true;
44         popout(u);
45     }
46 }
47 void init() {
48     cin >> n >> m;
49     fill(low, low+maxn, INF);
50     REP(i, m) {
51         int u, v;
52         cin >> u >> v;
53         g[u].emplace_back(i);

```

```

54     g[v].emplace_back(i);
55     E.emplace_back(u^v);
56 }
57 }
58 void solve() {
59     FOR(i, 1, n+1, 1) {
60         if (!dfn[i]) dfs(i, true);
61     }
62     vector<int> ans;
63     int cnt = 0;
64     FOR(i, 1, n+1, 1) {
65         if (isap[i]) cnt++, ans.emplace_back(i);
66     }
67     cout << cnt << endl;
68     Each(i, ans) cout << i << ' ';
69     cout << endl;
70 }

```

7.6 BCC - Bridge

```

1  int n, m;
2  vector<int> g[maxn], E;
3  int low[maxn], dfn[maxn], instp;
4  int bccnt, bccid[maxn];
5  stack<int> stk;
6  bitset<maxm> vis, isbrg;
7  void init() {
8      cin >> n >> m;
9      REP(i, m) {
10         int u, v;
11         cin >> u >> v;
12         E.emplace_back(u^v);
13         g[u].emplace_back(i);
14         g[v].emplace_back(i);
15     }
16     fill(low, low+maxn, INF);
17 }
18 void popout(int u) {
19     bccnt++;
20     while (!stk.empty()) {
21         int v = stk.top();
22         if (v == u) break;
23         stk.pop();
24         bccid[v] = bccnt;
25     }
26 }
27 void dfs(int u) {
28     stk.push(u);
29     low[u] = dfn[u] = ++instp;
30
31     Each(e, g[u]) {
32         if (vis[e]) continue;
33         vis[e] = true;
34
35         int v = E[e]^u;
36         if (dfn[v]) {
37             // back edge
38             low[u] = min(low[u], dfn[v]);
39         } else {
40             // tree edge
41             dfs(v);
42             low[u] = min(low[u], low[v]);
43             if (low[v] == dfn[v]) {
44                 isbrg[e] = true;
45                 popout(u);

```

```

46     }
47 }
48 }
49 }
50 void solve() {
51     FOR(i, 1, n+1, 1) {
52         if (!dfn[i]) dfs(i);
53     }
54     vector<pii> ans;
55     vis.reset();
56     FOR(u, 1, n+1, 1) {
57         Each(e, g[u]) {
58             if (!isbrg[e] || vis[e]) continue;
59             vis[e] = true;
60             int v = E[e]^u;
61             ans.emplace_back(mp(u, v));
62         }
63     }
64     cout << (int)ans.size() << endl;
65     Each(e, ans) cout << e.F << ' ' << e.S << endl;
66 }

```

7.7 SCC - Tarjan with 2-SAT

```

1  // Author: Ian
2  // 2-sat + tarjan SCC
3  void solve() {
4      int n, r, l; cin >> n >> r >> l;
5      V<P<int>> v(l);
6      for (auto& [a, b] : v)
7          cin >> a >> b;
8      V<V<int>> e(2 * l);
9      for (int i = 0; i < l; i++)
10         for (int j = i + 1; j < l; j++) {
11             if (v[i].first == v[j].first &&
12                 abs(v[i].second - v[j].second) <= 2 * r) {
13                 e[i << 1].emplace_back(j << 1 | 1);
14                 e[j << 1].emplace_back(i << 1 | 1);
15             }
16             if (v[i].second == v[j].second &&
17                 abs(v[i].first - v[j].first) <= 2 * r) {
18                 e[i << 1 | 1].emplace_back(j << 1);
19                 e[j << 1 | 1].emplace_back(i << 1);
20             }
21         }
22     V<bool> ins(2 * l, false);
23     V<int> scc(2 * l), dfn(2 * l, -1), low(2 * l, inf);
24     stack<int> s;
25     function<void(int)> dfs = [&](int x) {
26         if (~dfn[x]) return;
27         static int t = 0;
28         dfn[x] = low[x] = t++;
29         s.push(x), ins[x] = true;
30         for (auto i : e[x])
31             if (dfs(i), ins[i])
32                 low[x] = min(low[x], low[i]);
33         if (dfn[x] == low[x]) {
34             static int ncnt = 0;
35             int p; do {
36                 ins[p = s.top()] = false;
37                 s.pop(), scc[p] = ncnt;
38             } while (p != x); ncnt++;
39         }
40     };
41     for (int i = 0; i < 2 * l; i++)

```

```

40     dfs(i);
41     for (int i = 0; i < l; i++)
42         if (scc[i << 1] == scc[i << 1 | 1]) {
43             cout << "NO" << endl;
44             return;
45         }
46     cout << "YES" << endl;
47 }

```

7.8 Eulerian Path - Undir

```

1 // Author: Gino
2 // Usage: build deg, G first, then eulerian()
3 int n, m; // number of vertices and edges
4 vector<int> deg; // degree
5 vector<set<pii>> G; // G[u] := {(v, edge id)}
6
7 vector<int> path_u, path_e;
8 void dfs(int u) {
9     while (!G[u].empty()) {
10         auto it = G[u].begin();
11         auto [v, i] = *it; G[u].erase(it);
12         G[v].erase(make_pair(u, i)); dfs(v);
13         path_u.emplace_back(v);
14         path_e.emplace_back(i);
15     }
16 }
17 void gogo(int s) {
18     path_u.clear(); path_e.clear();
19     dfs(s); path_u.emplace_back(s);
20     reverse(path_u.begin(), path_u.end());
21     reverse(path_e.begin(), path_e.end());
22 }
23 bool eulerian() {
24     int oddcnt = 0, s = -1;
25     for (int u = 1; u <= n; u++)
26         if (deg[u] & 1)
27             oddcnt++, s = u;
28
29     if (oddcnt != 0 && oddcnt != 2) return false;
30     if (s == -1) {
31         s = 1; for (int u = 1; u <= n; u++)
32             if (deg[u] > 0)
33                 s = u;
34     }
35     gogo(s);
36
37     for (int u = 1; u <= n; u++)
38         if ((int)G[u].size() > 0)
39             return false;
40     return true;
41 }

```

7.9 Eulerian Path - Dir

```

1 // Author: Gino
2 // Usage: build ind, oud, G first, then eulerian()
3 int n, m; // number of vertices, edges
4 vector<int> ind, oud; // indegree, outdegree
5 vector<vector<pii>> G; // G[u] := {(v, edge id)}
6
7 vector<int> path_u, path_e;
8 void dfs(int u) {
9     while (!G[u].empty()) {
10         auto [v, i] = G[u].back(); G[u].pop_back();
11         dfs(v);
12         path_u.emplace_back(v);

```

```

13         path_e.emplace_back(i);
14     }
15 }
16 void gogo(int s) {
17     path_u.clear(); path_e.clear();
18     dfs(s); path_u.emplace_back(s);
19     reverse(path_u.begin(), path_u.end());
20     reverse(path_e.begin(), path_e.end());
21 }
22 bool eulerian() {
23     int s = -1;
24     for (int u = 1; u <= n; u++) {
25         if (abs(oud[u] - ind[u]) > 1) return false;
26         if (oud[u] - ind[u] == 1) {
27             if (s != -1) return false;
28             s = u;
29         }
30     }
31     if (s == -1) {
32         s = 1; for (int u = 1; u <= n; u++)
33             if (ind[u] > 0)
34                 s = u;
35     }
36     gogo(s);
37     for (int u = 1; u <= n; u++)
38         if ((int)G[u].size() > 0)
39             return false;
40
41     return true;
42 }

```

7.10 Kth Shortest Path

```

1 // time: O(|E| \lg |E| + |V| \lg |V| + K)
2 // memory: O(|E| \lg |E| + |V|)
3 struct KSP{ // 1-base
4     struct nd{
5         int u,v; ll d;
6         nd(int ui=0,int vi=0,ll di=INF){ u=ui; v=vi;
7             d=di; }
8     };
9     struct heap{ nd* edge; int dep; heap* chd[4]; };
10     static int cmp(heap* a,heap* b)
11     { return a->edge->d > b->edge->d; }
12     struct node{
13         int v; ll d; heap* H; nd* E;
14         node(){ }
15         node(ll _d,int _v,nd* _E){ d=_d; v=_v; E=_E; }
16         node(heap* _H,ll _d){ H=_H; d=_d; }
17         friend bool operator<(node a,node b)
18         { return a.d > b.d; }
19     };
20     int n,k,s,t,dst[N]; nd *nxt[N];
21     vector<nd*> g[N],rg[N]; heap *nullNd,*head[N];
22     void init(int _n,int _k,int _s,int _t){
23         n=_n; k=_k; s=_s; t=_t;
24         for(int i=1;i<=n;i++){
25             g[i].clear(); rg[i].clear();
26             nxt[i]=NULL; head[i]=NULL; dst[i]=-1;
27         }
28     }
29     void addEdge(int ui,int vi,ll di){
30         nd* e=new nd(ui,vi,di);
31         g[ui].push_back(e); rg[vi].push_back(e);

```

```

32 queue<int> dfsQ;
33 void dijkstra(){
34     while(dfsQ.size()) dfsQ.pop();
35     priority_queue<node> Q; Q.push(node(0,t,NULL));
36     while (!Q.empty()){
37         node p=Q.top(); Q.pop(); if(dst[p.v]!
38         ==-1)continue;
39         dst[p.v]=p.d; nxt[p.v]=p.E; dfsQ.push(p.v);
40         for(auto e:rg[p.v]) Q.push(node(p.d+e->d,e-
41         >u,e));
42     }
43     heap* merge(heap* curNd,heap* newNd){
44         if(curNd==nullNd) return newNd;
45         heap* root=new
46         heap;memcpy(root,curNd,sizeof(heap));
47         if(newNd->edge->d<curNd->edge->d){
48             root->edge=newNd->edge;
49             root->chd[2]=newNd->chd[2];
50             root->chd[3]=newNd->chd[3];
51             newNd->edge=curNd->edge;
52             newNd->chd[2]=curNd->chd[2];
53             newNd->chd[3]=curNd->chd[3];
54         }
55         if(root->chd[0]->dep<root->chd[1]->dep)
56             root->chd[0]=merge(root->chd[0],newNd);
57         else root->chd[1]=merge(root->chd[1],newNd);
58         root->dep=max(root->chd[0]->dep,
59         root->chd[1]->dep)+1;
60         return root;
61     }
62     vector<heap*> V;
63     void build(){
64         nullNd=new heap; nullNd->dep=0; nullNd-
65         >edge=new nd;
66         fill(nullNd->chd,nullNd->chd+4,nullNd);
67         while(not dfsQ.empty()){
68             int u=dfsQ.front(); dfsQ.pop();
69             if(!nxt[u]) head[u]=nullNd;
70             else head[u]=head[nxt[u]->v];
71             V.clear();
72             for(auto&& e:g[u]){
73                 int v=e->v;
74                 if(dst[v]==-1) continue;
75                 e->d+=dst[v]-dst[u];
76                 if(nxt[u]!=e){
77                     heap* p=new heap;fill(p->chd,p-
78                     >chd+4,nullNd);
79                     p->dep=1; p->edge=e; V.push_back(p);
80                 }
81             }
82             if(V.empty()) continue;
83             make_heap(V.begin(),V.end(),cmp);
84             #define L(X) ((X<<1)+1)
85             #define R(X) ((X<<1)+2)
86             for(size_t i=0;i<V.size();i++){
87                 if(L(i)<V.size()) V[i]->chd[2]=V[L(i)];
88                 else V[i]->chd[2]=nullNd;
89                 if(R(i)<V.size()) V[i]->chd[3]=V[R(i)];
90                 else V[i]->chd[3]=nullNd;
91             }
92             head[u]=merge(head[u],V.front());
93         }
94     }
95     vector<ll> ans;

```

```

92 void first_K(){
93     ans.clear(); priority_queue<node> Q;
94     if(dst[s]==-1) return;
95     ans.push_back(dst[s]);
96     if(head[s]!=nullNd)
97         Q.push(node(head[s],dst[s]+head[s]->edge-
98         >d));
99     for(int _=1;_<k and not Q.empty();_++){
100         node p=Q.top(); q=Q.pop();
101         ans.push_back(p.d);
102         if(head[p.H->edge->v]!=nullNd){
103             q.H=head[p.H->edge->v]; q.d=p.d+q.H->edge-
104             >d;
105             Q.push(q);
106         }
107         for(int i=0;i<4;i++){
108             if(p.H->chd[i]!=nullNd){
109                 q.H=p.H->chd[i];
110                 q.d=p.d-p.H->edge->d+p.H->chd[i]->edge-
111                 >d;
112                 Q.push(q);
113             }
114         }
115     }
116     void solve(){ // ans[i] stores the i-th shortest
117         path
118         dijkstra(); build();
119         first_K(); // ans.size() might less than k
120     }
121     solver;

```

7.11 System of Difference Constraints

```

1 vector<vector<pair<int, ll>>> G;
2 void add(int u, int v, ll w) {
3     G[u].emplace_back(make_pair(v, w));
4 }

```

- $x_u - x_v \leq c \Rightarrow \text{add}(v, u, c)$
- $x_u - x_v \geq c \Rightarrow \text{add}(u, v, -c)$
- $x_u - x_v = c \Rightarrow \text{add}(v, u, c), \text{add}(u, v, -c)$
- $x_u \geq c \Rightarrow \text{add super vertex } x_0 = 0, \text{ then } x_u - x_0 \geq c \Rightarrow \text{add}(u, 0, -c)$
- Don't forget non-negative constraints for every variable if specified implicitly.
- Interval sum $\Rightarrow$ Use prefix sum to transform into differential constraints. Don't forget $S_{i+1} - S_i \geq 0$ if x_i needs to be non-negative.
- $\frac{x_u}{x_v} \leq c \Rightarrow \log x_u - \log x_v \leq \log c$

8 String

8.1 Rolling Hash

```

1 const ll C = 27;
2 inline int id(char c) {return c-'a'+1;}
3 struct RollingHash {
4     string s; int n; ll mod;
5     vector<ll> Cexp, hs;
6     RollingHash(string& _s, ll _mod):
7         s(_s), n((int)_s.size()), mod(_mod)
8     {
9         Cexp.assign(n, 0);
10        hs.assign(n, 0);
11        Cexp[0] = 1;
12        for (int i = 1; i < n; i++) {
13            Cexp[i] = Cexp[i-1] * C;
14            if (Cexp[i] >= mod) Cexp[i] %= mod;
15        }
16        hs[0] = id(s[0]);

```

```

17     for (int i = 1; i < n; i++) {
18         hs[i] = hs[i-1] * C + id(s[i]);
19         if (hs[i] >= mod) hs[i] %= mod;
20     }
21     inline ll query(int l, int r) {
22         ll res = hs[r] - (l ? hs[l-1] * Cexp[r-
23             l+1] : 0);
24         res = (res % mod + mod) % mod;
25         return res; }
26 };

```

8.2 Trie

```

1 struct node {
2     int c[26]; ll cnt;
3     node(): cnt(0) {memset(c, 0, sizeof(c));}
4     node(ll x): cnt(x) {memset(c, 0, sizeof(c));}
5 };
6 struct Trie {
7     vector<node> t;
8     void init() {
9         t.clear();
10        t.emplace_back(node());
11    }
12    void insert(string s) { int ptr = 0;
13        for (auto& i : s) {
14            if (!t[ptr].c[i-'a']) {
15                t.emplace_back(node());
16                t[ptr].c[i-'a'] =
17                    (int)t.size()-1; }
18            ptr = t[ptr].c[i-'a']; }
19        t[ptr].cnt++; }
20 } trie;

```

8.3 KMP

```

1 int n, m;
2 string s, p;
3 vector<int> f;
4 void build() {
5     f.clear(); f.resize(m, 0);
6     int ptr = 0; for (int i = 1; i < m; i++) {
7         while (ptr && p[i] != p[ptr]) ptr = f[ptr-1];
8         if (p[i] == p[ptr]) ptr++;
9         f[i] = ptr;
10    }
11    void init() {
12        cin >> s >> p;
13        n = (int)s.size();
14        m = (int)p.size();
15        build(); }
16    void solve() {
17        int ans = 0, pi = 0;
18        for (int si = 0; si < n; si++) {
19            while (pi && s[si] != p[pi]) pi = f[pi-1];
20            if (s[si] == p[pi]) pi++;
21            if (pi == m) ans++, pi = f[pi-1];
22        }
23        cout << ans << endl; }

```

8.4 Z Value

```

1 string is, it, s;
2 int n; vector<int> z;
3 void init() {
4     cin >> is >> it;
5     s = it+'0'+is;

```

```

6     n = (int)s.size();
7     z.resize(n, 0); }
8 void solve() {
9     int ans = 0; z[0] = n;
10    for (int i = 1, l = 0, r = 0; i < n; i++) {
11        if (i <= r) z[i] = min(z[i-l], r-i+1);
12        while (i+z[i] < n && s[z[i]] == s[i+z[i]])
13            z[i]++;
14        if (i+z[i]-1 > r) l = i, r = i+z[i]-1;
15        if (z[i] == (int)it.size()) ans++;
16    }
17    cout << ans << endl; }

```

8.5 Manacher

```

1 int n; string S, s;
2 vector<int> m;
3 void manacher() {
4     s.clear(); s.resize(2*n+1, '.');
5     for (int i = 0, j = 1; i < n; i++, j += 2) s[j] =
6         S[i];
7     m.clear(); m.resize(2*n+1, 0);
8     // m[i] := max k such that s[i-k, i+k] is
9     // palindrome
10    int mx = 0, mxk = 0;
11    for (int i = 1; i < 2*n+1; i++) {
12        if (mx-(i-mx) >= 0) m[i] = min(m[mx-(i-mx)],
13            mx+mxk-i);
14        while (0 <= i-m[i]-1 && i+m[i]+1 < 2*n+1 &&
15            s[i-m[i]-1] == s[i+m[i]+1]) m[i]++;
16        if (i+m[i] > mx+mxk) mx = i, mxk = m[i];
17    }
18    void init() { cin >> S; n = (int)S.size(); }
19    void solve() {
20        manacher();
21        int mx = 0, ptr = 0;
22        for (int i = 0; i < 2*n+1; i++) if (mx < m[i])
23            { mx = m[i]; ptr = i; }
24        for (int i = ptr-mx; i <= ptr+mx; i++)
25            if (s[i] != '.') cout << s[i];
26        cout << endl; }

```

8.6 Suffix Array

```

1 #define F first
2 #define S second
3 struct SuffixArray { // don't forget s += "$";
4     int n; string s;
5     vector<int> suf, lcp, rk;
6     vector<int> cnt, pos;
7     vector<pair<pii, int>> buc[2];
8     void init(string _s) {
9         s = _s; n = (int)s.size();
10        // resize(n): suf, rk, cnt, pos, lcp, buc[0~1]
11    }
12    void radix_sort() {
13        for (int t : {0, 1}) {
14            fill(cnt.begin(), cnt.end(), 0);
15            for (auto& i : buc[t]) cnt[ (t ?
16                i.F.F : i.F.S) ]++;
17            for (int i = 0; i < n; i++)
18                pos[i] = (!i ? 0 : pos[i-1] +
19                    cnt[i-1]);
20            for (auto& i : buc[t])
21                buc[t^1][pos[ (t ? i.F.F :
22                    i.F.S) ]++] = i;
23        }
24        bool fill_suf() {

```



```

22     bool end = true;
23     for (int i = 0; i < n; i++) suf[i] = buc[0]
[i].S;
24     rk[suf[0]] = 0;
25     for (int i = 1; i < n; i++) {
26         int dif = (buc[0][i].F != buc[0]
[i-1].F);
27         end &= dif;
28         rk[suf[i]] = rk[suf[i-1]] + dif;
29     } return end;
30 }
31 void sa() {
32     for (int i = 0; i < n; i++)
33         buc[0][i] = make_pair(make_pair(s[i],
s[i]), i);
34     sort(buc[0].begin(), buc[0].end());
35     if (fill_suf()) return;
36     for (int k = 0; (1<<k) < n; k++) {
37         for (int i = 0; i < n; i++)
38             buc[0][i] =
make_pair(make_pair(rk[i], rk[(i + (1<<k)) % n]),
i);
39         radix_sort();
40         if (fill_suf()) return;
41     }
42     void LCP() { int k = 0;
43         for (int i = 0; i < n-1; i++) {
44             if (rk[i] == 0) continue;
45             int pi = rk[i];
46             int j = suf[pi-1];
47             while (i+k < n && j+k < n && s[i+k] ==
s[j+k]) k++;
48             lcp[pi] = k;
49             k = max(k-1, 0);
50         }
51 };
52 SuffixArray suffixarray;

```

8.7 Suffix Automaton

```

1 // any path start from root forms a substring of S
2 // occurrence of P: iff SAM can run on input word P
3 // number of different substring: ds[1]-1
4 // total length of all different substring: dsl[1]
5 // max/min length of state i: mx[i]/mx[mom[i]]+1
6 // assume a run on input word P end at state i:
7 // number of occurrences of P: cnt[i]
8 // first occurrence position of P: fp[i]-|P|+1
9 // all position: !clone nodes in dfs from i through
rrom
10 const int MXM=1000010;
11 struct SAM{
12     int tot, root, lst, mom[MXM], mx[MXM]; // ind[MXM]
13     int nxt[MXM][33]; //
cnt[MXM], ds[MXM], dsl[MXM], fp[MXM]
14     // bool v[MXM], clone[MXN]
15     int newNode(){
16         int res=++tot; fill(nxt[res], nxt[res]+33, 0);
17         mom[res]=mx[res]=0; // cnt=ds=dsl=fp=v=clone=0
18         return res;
19     }
20     void init(){ tot=0; root=newNode(); lst=root; }
21     void push(int c){
22         int p=lst, np=newNode(); //
cnt[np]=1, clone[np]=0
23         mx[np]=mx[p]+1; // fp[np]=mx[np]-1
24         for(; p&&nxt[p][c]==0; p=mom[p]) nxt[p][c]=np;

```

```

25         if(p==0) mom[np]=root;
26         else{
27             int q=nxt[p][c];
28             if(mx[p]+1==mx[q]) mom[np]=q;
29             else{
30                 int nq=newNode(); //
fp[nq]=fp[q], clone[nq]=1
31                 mx[nq]=mx[p]+1;
32                 for(int i=0; i<33; i++) nxt[nq][i]=nxt[q][i];
33                 mom[nq]=mom[q]; mom[q]=nq; mom[np]=nq;
34                 for(; p&&nxt[p][c]==q; p=mom[p]) nxt[p]
[c]=nq;
35             }
36         }
37         lst=np;
38     }
39     void calc(){
40         calc(root); iota(ind, ind+tot, 1);
41         sort(ind, ind+tot, [&](int i, int j){return
mx[i]<mx[j];});
42         for(int i=tot-1; i>=0; i--)
43             cnt[mom[ind[i]]]+=cnt[ind[i]];
44     }
45     void calc(int x){
46         v[x]=ds[x]=1, dsl[x]=0; //
rmom[mom[x]].push_back(x);
47         for(int i=0; i<26; i++){
48             if(nxt[x][i]){
49                 if(!v[nxt[x][i]]) calc(nxt[x][i]);
50                 ds[x]+=ds[nxt[x][i]];
51                 dsl[x]+=ds[nxt[x][i]]+dsl[nxt[x][i]];
52             } }
53     void push(char *str){
54         for(int i=0; str[i]; i++) push(str[i]-'a');
55     }
56 } sam;

```

8.8 SA-IS

```

1 const int N=300010;
2 struct SA{
3     #define REP(i,n) for(int i=0; i<int(n); i++)
4     #define REPl(i,a,b) for(int i=(a); i<=int(b); i++)
5     bool _t[N*2]; int _s[N*2], _sa[N*2];
6     int _c[N*2], x[N], _p[N], _q[N*2], hei[N], r[N];
7     int operator [](int i){ return _sa[i]; }
8     void build(int *s, int n, int m){
9         memcpy(_s, s, sizeof(int)*n);
10        sais(_s, _sa, _p, _q, _t, _c, n, m); mkhei(n);
11    }
12    void mkhei(int n){
13        REP(i,n) r[_sa[i]]=i;
14        hei[0]=0;
15        REP(i,n) if(r[i]) {
16            int ans=i>0?max(hei[r[i-1]]-1, 0):0;
17            while(_s[i+ans]==_s[_sa[r[i]-1]+ans]) ans++;
18            hei[r[i]]=ans;
19        }
20    }
21    void sais(int *s, int *sa, int *p, int *q, bool
*t, int *c, int n, int z){
22        bool uniq=t[n-1]=true, neq;
23        int nn=0, nmz=-1, *nsa=sa+n, *ns=s+n, lst=-1;
24        #define MS0(x,n) memset((x), 0, n*sizeof(*(x)))
25        #define MAGIC(XD) MS0(sa, n); \
memcpy(x, c, sizeof(int)*z); XD; \

```

```

27 memcpy(x+1,c,sizeof(int)*(z-1));\
28 REP(i,n) if(sa[i]&&!t[sa[i]-1]) sa[x[s[sa[i]-1]]+
+=sa[i]-1];\
29 memcpy(x,c,sizeof(int)*z);\
30 for(int i=n-1;i>=0;i--) if(sa[i]&&t[sa[i]-1]) sa[--
x[s[sa[i]-1]]]=sa[i]-1;
31 MS0(c,z); REP(i,n) uniq&=++c[s[i]]<2;
32 REP(i,z-1) c[i+1]+=c[i];
33 if(uniq) { REP(i,n) sa[--c[s[i]]]=i; return; }
34 for(int i=n-2;i>=0;i--)
35     t[i]=(s[i]==s[i+1]?t[i+1]:s[i]<s[i+1]);
36 MAGIC(REP1(i,1,n-1) if(t[i]&&!t[i-1]) sa[--
x[s[i]]]=p[q[i]=nn++]=i);
37 REP(i,n) if(sa[i]&&t[sa[i]]&&!t[sa[i]-1]){
38     neq=lst<0||memcmp(s+sa[i],s+lst,
(p[q[sa[i]]+1]-sa[i])*sizeof(int));
39     ns[q[lst=sa[i]]]=nmxz+neq;
40 }
41 sais(ns,nsa,p+nn,q+n,t+n,c+z,nn,nmxz+1);
42 MAGIC(for(int i=nn-1;i>=0;i--) sa[--
x[s[p[nsa[i]]]]=p[nsa[i]]];
43 }
44 }sa;
45 int H[N],SA[N],RA[N];
46 void suffix_array(int* ip,int len){
47     // should padding a zero in the back
48     // ip is int array, len is array length
49     // ip[0..n-1] != 0, and ip[len]=0
50     ip[len++]=0; sa.build(ip,len,128);
51     memcpy(H,sa.hei+1,len<<2);
52     memcpy(SA,sa._sa+1,len<<2);
53     for(int i=0;i<len;i++) RA[i]=sa.r[i]-1;
54     // resulting height, sa array \in [0,len)
55 }

```

8.9 Minimum Rotation

```

1 // lexicographically minimal rotation
2 // rotate(begin(s), begin(s)+minRotation(s),
   end(s))
3 int minRotation(string s) {
4     int a = 0, n = s.size(); s += s;
5     for(int b = 0; b < n; b++) for(int k = 0; k < n; k+
+=) {
6         if(a + k == b || s[a + k] < s[b + k]) {
7             b += max(0, k - 1);
8             break; }
9         if(s[a + k] > s[b + k]) {
10             a = b;
11             break;
12         } }
13     return a; }

```

8.10 Aho Corasick

```

1 struct ACautomata{
2     struct Node{
3         int cnt;
4         Node *go[26], *fail, *dic;
5         Node(){
6             cnt = 0; fail = 0; dic=0;
7             memset(go,0,sizeof(go));
8         }
9     }pool[1048576],*root;
10     int nMem;
11     Node* new_Node(){
12         pool[nMem] = Node();
13         return &pool[nMem++];

```

```

14     }
15     void init() { nMem = 0; root = new_Node(); }
16     void add(const string &str)
17     { insert(root,str,0); }
18     void insert(Node *cur, const string &str, int
pos){
19         for(int i=pos;i<str.size();i++){
20             if(!cur->go[str[i]-'a'])
21                 cur->go[str[i]-'a'] = new_Node();
22             cur=cur->go[str[i]-'a'];
23         }
24         cur->cnt++;
25     }
26     void make_fail(){
27         queue<Node*> que;
28         que.push(root);
29         while (!que.empty()){
30             Node* fr=que.front(); que.pop();
31             for (int i=0; i<26; i++){
32                 if (fr->go[i]){
33                     Node *ptr = fr->fail;
34                     while (ptr && !ptr->go[i]) ptr = ptr-
>fail;
35                     fr->go[i]->fail=ptr=(ptr?ptr-
>go[i]:root);
36                     fr->go[i]->dic=(ptr->cnt?ptr:ptr->dic);
37                     que.push(fr->go[i]);
38                 } } } }
39 }AC;

```

9 Geometry

9.1 Template

```

1 // Author: Gino
2 typedef long long T;
3 // typedef long double T;
4 const long double eps = 1e-8;
5
6 short sgn(T x) {
7     if (abs(x) < eps) return 0;
8     return x < 0 ? -1 : 1;
9 }
10
11 struct Pt {
12     T x, y;
13     Pt(T _x=0, T _y=0):x(_x), y(_y) {}
14     Pt operator+(Pt a) { return Pt(x+a.x, y+a.y); }
15     Pt operator-(Pt a) { return Pt(x-a.x, y-a.y); }
16     Pt operator*(T a) { return Pt(x*a, y*a); }
17     Pt operator/(T a) { return Pt(x/a, y/a); }
18     T operator*(Pt a) { return x*a.x + y*a.y; }
19     T operator^(Pt a) { return x*a.y - y*a.x; } // 不
   要打反
20     bool operator<(Pt a)
21     { return x < a.x || (x == a.x && y < a.y); }
22     //return sgn(x-a.x) < 0 || (sgn(x-a.x) == 0 &&
   sgn(y-a.y) < 0); }
23     bool operator==(Pt a)
24     { return sgn(x-a.x) == 0 && sgn(y-a.y) == 0; }
25 };
26
27 Pt mv(Pt a, Pt b) { return b-a; }
28 T len2(Pt a) { return a*a; }
29 T dis2(Pt a, Pt b) { return len2(b-a); }
30

```

```

31 short ori(Pt a, Pt b) { return ((a^b)>0) -
    ((a^b)<0); }
32 bool onseg(Pt p, Pt l1, Pt l2) {
33     Pt a = mv(p, l1), b = mv(p, l2);
34     return ((a^b) == 0) && ((a*b) <= 0);
35 }

```

9.2 Basic Operations

```

1 // Author: Gino
2 // Function: Check if a point P sits in a polygon
  (doesn't have to be convex hull)
3 // 0 = Bound, 1 = In, -1 = Out
4 short inPoly(Pt p) {
5     for (int i = 0; i < n; i++)
6         if (onseg(p, E[i], E[(i+1)%n])) return 0;
7     int cnt = 0;
8     for (int i = 0; i < n; i++)
9         if (banana(p, Pt(p.x+1, p.y+2e9), E[i],
10             E[(i+1)%n]))
11             cnt ^= 1;
12     return (cnt ? 1 : -1);
13 }

1 // Author: Gino
2 int ud(Pt a) { // up or down half plane
3     if (a.y > 0) return 0;
4     if (a.y < 0) return 1;
5     return (a.x >= 0 ? 0 : 1);
6 }
7 sort(ALL(E), [&](const Pt& a, const Pt& b){
8     if (ud(a) != ud(b)) return ud(a) < ud(b);
9     return (a^b) > 0;
10 });

1 // Author: Gino
2 // Function: check if (p1---p2) (q1---q2) banana
3 inline bool banana(Pt p1, Pt p2, Pt q1, Pt q2) {
4     if (onseg(p1, q1, q2) || onseg(p2, q1, q2) ||
5         onseg(q1, p1, p2) || onseg(q2, p1, p2)) {
6         return true;
7     }
8     Pt p = mv(p1, p2), q = mv(q1, q2);
9     return (ori(p, mv(p1, q1)) * ori(p, mv(p1, q2)) < 0
10         &&
11         ori(q, mv(q1, p1)) * ori(q, mv(q1, p2)) <
12         0);
13 }

1 // Author: Gino
2 // T: long double
3 Pt bananaPoint(Pt p1, Pt p2, Pt q1, Pt q2) {
4     if (onseg(q1, p1, p2)) return q1;
5     if (onseg(q2, p1, p2)) return q2;
6     if (onseg(p1, q1, q2)) return p1;
7     if (onseg(p2, q1, q2)) return p2;
8     double s = abs(mv(p1, p2) ^ mv(p1, q1));
9     double t = abs(mv(p1, p2) ^ mv(p1, q2));
10    return q2 * (s/(s+t)) + q1 * (t/(s+t));
11 }

1 // Author: Gino
2 vector<Pt> hull;
3 void convexHull() {
4     hull.clear(); sort(E.begin(), E.end());
5     for (int t : {0, 1}) {
6         int b = (int)hull.size();
7         for (auto& ei : E) {

```

```

8             while ((int)hull.size() - b >= 2 &&
9                 ori(mv(hull[(int)hull.size()-2],
10                     hull.back()),
11                     mv(hull[(int)hull.size()-2],
12                         ei)) == -1) {
13                 hull.pop_back();
14             }
15             hull.emplace_back(ei);
16         }
17     }
18 }

```

```

1 // Author: Gino
2 // Function: Return doubled area of a polygon
3 T dbarea(vector<Pt>& e) {
4     ll res = 0;
5     for (int i = 0; i < (int)e.size(); i++)
6         res += e[i]^e[(i+1)%SZ(e)];
7     return abs(res);
8 }

```

9.3 Lower Concave Hull

```

1 // Author: Unknown
2 struct Line {
3     mutable ll m, b, p;
4     bool operator<(const Line& o) const { return m <
5         o.m; }
6     bool operator<(ll x) const { return p < x; }
7 };
8 struct LineContainer : multiset<Line, less<>> {
9     // (for doubles, use inf = 1/.0, div(a,b) = a/b)
10    const ll inf = LLONG_MAX;
11    ll div(ll a, ll b) { // floored division
12        return a / b - ((a ^ b) < 0 && a % b); }
13    bool isect(iterator x, iterator y) {
14        if (y == end()) { x->p = inf; return false; }
15        if (x->m == y->m) x->p = x->b > y->b ? inf : -
16            inf;
17        else x->p = div(y->b - x->b, x->m - y->m);
18        return x->p >= y->p;
19    }
20    void add(ll m, ll b) {
21        auto z = insert({m, b, 0}), y = z++, x = y;
22        while (isect(y, z)) z = erase(z);
23        if (x != begin() && isect(--x, y)) isect(x, y =
24            erase(y));
25        while ((y = x) != begin() && (--x)->p >= y->p)
26            isect(x, erase(y));
27    }
28    ll query(ll x) {
29        assert(!empty());
30        auto l = *lower_bound(x);
31        return l.m * x + l.b;
32    }
33 };

```

9.4 Pick's Theorem

Consider a polygon which vertices are all lattice points.

Let i = number of points inside the polygon.

Let b = number of points on the boundary of the polygon.

Then we have the following formula:

$$\text{Area} = i + \frac{b}{2} - 1$$

9.5 Minimum Enclosing Circle

```

1 // Author: Gino
2 // Function: Find Min Enclosing Circle using
  Randomized O(n) Algorithm
3 Pt circumcenter(Pt A, Pt B, Pt C) {
4 // a1(x-A.x) + b1(y-A.y) = c1
5 // a2(x-A.x) + b2(y-A.y) = c2
6 // solve using Cramer's rule
7 T a1 = B.x-A.x, b1 = B.y-A.y, c1 = dis2(A, B)/2.0;
8 T a2 = C.x-A.x, b2 = C.y-A.y, c2 = dis2(A, C)/2.0;
9 T D = Pt(a1, b1) ^ Pt(a2, b2);
10 T Dx = Pt(c1, b1) ^ Pt(c2, b2);
11 T Dy = Pt(a1, c1) ^ Pt(a2, c2);
12 if (D == 0) return Pt(-INF, -INF);
13 return A + Pt(Dx/D, Dy/D);
14 }
15 Pt center; T r2;
16
17 void minEncloseCircle() {
18 mt19937
19 gen(chrono::steady_clock::now().time_since_epoch().count());
20 shuffle(ALL(E), gen);
21 center = E[0], r2 = 0;
22 for (int i = 0; i < n; i++) {
23     if (dis2(center, E[i]) <= r2) continue;
24     center = E[i], r2 = 0;
25     for (int j = 0; j < i; j++) {
26         if (dis2(center, E[j]) <= r2) continue;
27         center = (E[i] + E[j]) / 2.0;
28         r2 = dis2(center, E[i]);
29         for (int k = 0; k < j; k++) {
30             if (dis2(center, E[k]) <= r2) continue;
31             center = circumcenter(E[i], E[j],
32                                   E[k]);
33             r2 = dis2(center, E[i]);
34         }
35     }
36 }

```

9.6 PolyUnion

```

1 // Author: Unknown
2 struct PY{
3     int n; Pt pt[5]; double area;
4     Pt& operator[](const int x){ return pt[x]; }
5     void init(){ //n,pt[0~n-1] must be filled
6         area=pt[n-1]^pt[0];
7         for(int i=0;i<n-1;i++) area+=pt[i]^pt[i+1];
8         if((area/=2)<0)reverse(pt,pt+n),area=-area;
9     }
10 };
11 PY py[500]; pair<double,int> c[5000];
12 inline double segP(Pt &p,Pt &p1,Pt &p2){
13     if(dcmp(p1.x-p2.x)==0) return (p.y-p1.y)/(p2.y-
14     p1.y);
15     return (p.x-p1.x)/(p2.x-p1.x);
16 }
17 double polyUnion(int n){ //py[0~n-1] must be filled
18     int i,j,ii,jj,ta,tb,r,d; double
19     z,w,s,sum=0,tc,td;
20     for(i=0;i<n;i++) py[i][py[i].n]=py[i][0];
21     for(i=0;i<n;i++){
22         for(ii=0;ii<py[i].n;ii++){

```

```

21         r=0;
22         c[r++]=make_pair(0.0,0); c[r+
23         +]=make_pair(1.0,0);
24         for(j=0;j<n;j++){
25             if(i==j) continue;
26             for(jj=0;jj<py[j].n;jj++){
27                 ta=dcmp(tri(py[i][ii],py[i][ii+1],py[j]
28                 [jj]));
29                 tb=dcmp(tri(py[i][ii],py[i][ii+1],py[j]
30                 [jj+1]));
31                 if(ta==0 && tb==0){
32                     if((py[j][jj+1]-py[j][jj])*(py[i]
33                     [ii+1]-py[i][ii])>0&&j<i){
34                         c[r++]=make_pair(segP(py[j][jj],py[i]
35                         [ii],py[i][ii+1]),1);
36                         c[r++]=make_pair(segP(py[j]
37                         [jj+1],py[i][ii],py[i][ii+1]),-1);
38                     }
39                     }else if(ta>=0 && tb<0){
40                         tc=tri(py[j][jj],py[j][jj+1],py[i]
41                         [ii]);
42                         td=tri(py[j][jj],py[j][jj+1],py[i]
43                         [ii+1]);
44                         c[r++]=make_pair(tc/(tc-td),1);
45                     }else if(ta<0 && tb>=0){
46                         tc=tri(py[j][jj],py[j][jj+1],py[i]
47                         [ii]);
48                         td=tri(py[j][jj],py[j][jj+1],py[i]
49                         [ii+1]);
50                         c[r++]=make_pair(tc/(tc-td),-1);
51                     } } }
52     sort(c,c+r);
53     z=min(max(c[0].first,0.0),1.0);
54     d=c[0].second; s=0;
55     for(j=1;j<r;j++){
56         w=min(max(c[j].first,0.0),1.0);
57         if(!d) s+=w-z;
58         d+=c[j].second; z=w;
59     }
60     sum+=(py[i][ii]^py[i][ii+1])*s;
61 }
62 return sum/2;
63 }

```

9.7 Minkowski Sum

```

1 // Author: Unknown
2 /* convex hull Minkowski Sum*/
3 #define INF 10000000000000LL
4 int pos( const Pt& tp ){
5     if( tp.Y == 0 ) return tp.X > 0 ? 0 : 1;
6     return tp.Y > 0 ? 0 : 1;
7 }
8 #define N 300030
9 Pt pt[ N ], qt[ N ], rt[ N ];
10 LL Lx,Rx;
11 int dn,un;
12 inline bool cmp( Pt a, Pt b ){
13     int pa=pos( a ),pb=pos( b );
14     if(pa==pb) return (a^b)>0;
15     return pa<pb;
16 }
17 int minkowskiSum(int n,int m){
18     int i,j,r,p,q,fi,fj;
19     for(i=1,p=0;i<n;i++){
20         if( pt[i].Y<pt[p].Y ||

```

```

21     (pt[i].Y==pt[p].Y && pt[i].X<pt[p].X) )
p=i; }
22 for(i=1,q=0;i<m;i++){
23     if( qt[i].Y<qt[q].Y ||
24         (qt[i].Y==qt[q].Y && qt[i].X<qt[q].X) )
q=i; }
25 rt[0]=pt[p]+qt[q];
26 r=1; i=p; j=q; fi=fj=0;
27 while(1){
28     if((fj&&j==q) ||
29         ( (!fi || i!=p) &&
30             cmp(pt[(p+1)%n]-pt[p],qt[(q+1)%m]-
qt[q]) ) ) ){
31         rt[r]=rt[r-1]+pt[(p+1)%n]-pt[p];
32         p=(p+1)%n;
33         fi=1;
34     }else{
35         rt[r]=rt[r-1]+qt[(q+1)%m]-qt[q];
36         q=(q+1)%m;
37         fj=1;
38     }
39     if(r<=1 || ((rt[r]-rt[r-1])^(rt[r-1]-rt[r-2]))!
=0) r++;
40     else rt[r-1]=rt[r];
41     if(i==p && j==q) break;
42 }
43 return r-1;
44 }
45 void initInConvex(int n){
46     int i,p,q;
47     LL Ly,Ry;
48     Lx=INF; Rx=-INF;
49     for(i=0;i<n;i++){
50         if(pt[i].X<Lx) Lx=pt[i].X;
51         if(pt[i].X>Rx) Rx=pt[i].X;
52     }
53     Ly=Ry=INF;
54     for(i=0;i<n;i++){
55         if(pt[i].X==Lx && pt[i].Y<Ly){ Ly=pt[i].Y;
p=i; }
56         if(pt[i].X==Rx && pt[i].Y>Ry){ Ry=pt[i].Y;
q=i; }
57     }
58     for(dn=0,i=p;i!=q;i=(i+1)%n){ qt[dn++]=pt[i]; }
59     qt[dn]=pt[q]; Ly=Ry=-INF;
60     for(i=0;i<n;i++){
61         if(pt[i].X==Lx && pt[i].Y>Ly){ Ly=pt[i].Y;
p=i; }
62         if(pt[i].X==Rx && pt[i].Y>Ry){ Ry=pt[i].Y;
q=i; }
63     }
64     for(un=0,i=p;i!=q;i=(i+n-1)%n){ rt[un++]=pt[i]; }
65     rt[un]=pt[q];
66 }
67 inline int inConvex(Pt p){
68     int L,R,M;
69     if(p.X<Lx || p.X>Rx) return 0;
70     L=0;R=dn;
71     while(L<R-1){ M=(L+R)/2;
72         if(p.X<qt[M].X) R=M; else L=M; }
73     if(tri(qt[L],qt[R],p)<0) return 0;
74     L=0;R=un;
75     while(L<R-1){ M=(L+R)/2;
76         if(p.X<rt[M].X) R=M; else L=M; }
77     if(tri(rt[L],rt[R],p)>0) return 0;
78     return 1;

```

```

79 }
80 int main(){
81     int n,m,i;
82     Pt p;
83     scanf("%d",&n);
84     for(i=0;i<n;i++){
85         scanf("%lld%lld",&pt[i].X,&pt[i].Y);
86         scanf("%d",&m);
87         for(i=0;i<m;i++){
88             scanf("%lld%lld",&qt[i].X,&qt[i].Y);
89             n=minkowskiSum(n,m);
90             for(i=0;i<n;i++) pt[i]=rt[i];
91             scanf("%d",&m);
92             for(i=0;i<m;i++){
93                 scanf("%lld%lld",&qt[i].X,&qt[i].Y);
94                 n=minkowskiSum(n,m);
95                 for(i=0;i<n;i++) pt[i]=rt[i];
96                 initInConvex(n);
97                 scanf("%d",&m);
98                 for(i=0;i<m;i++){
99                     scanf("%lld %lld",&p.X,&p.Y);
100                     p.X*=3; p.Y*=3;
101                     puts(inConvex(p)?"YES":"NO");
102                 }
103             }
104         }
105     }

```

10 Number Theory

10.1 Prime Sieve and Defactor

```

1 // Author: Gino
2 const int maxc = 1e6 + 1;
3 vector<int> lpf;
4 vector<int> prime;
5
6 void seive() {
7     prime.clear();
8     lpf.resize(maxc, 1);
9     for (int i = 2; i < maxc; i++) {
10         if (lpf[i] == 1) {
11             lpf[i] = i;
12             prime.emplace_back(i);
13         }
14         for (auto& j : prime) {
15             if (i * j >= maxc) break;
16             lpf[i * j] = j;
17             if (j == lpf[i]) break;
18         }
19     }
20     vector<pii> fac;
21     void defactor(int u) {
22         fac.clear();
23         while (u > 1) {
24             int d = lpf[u];
25             fac.emplace_back(make_pair(d, 0));
26             while (u % d == 0) {
27                 u /= d;
28                 fac.back().second++;
29             }
30         }
31     }

```

10.2 Harmonic Series

```

1 // Author: Gino
2 // O(n log n)
3 for (int i = 1; i <= n; i++) {
4     for (int j = i; j <= n; j += i) {
5         // O(1) code
6     }

```

```

7 }
8
9 // PIE
10 // given array a[0], a[1], ..., a[n - 1]
11 // calculate dp[x] = number of pairs (a[i], a[j])
   such that
12 //             gcd(a[i], a[j]) = x    // (i <
   j)
13 //
14 // idea: let mc(x) = # of y s.t. x|y
15 //             f(x) = # of pairs s.t. gcd(a[i],
   a[j]) >= x
16 //             f(x) = C(mc(x), 2)
17 //             dp[x] = f(x) - sum(dp[y], x < y and x|
   y)
18 const int maxc = 1e6;
19 vector<int> cnt(maxc + 1, 0), dp(maxc + 1, 0);
20 for (int i = 0; i < n; i++)
21     cnt[a[i]]++;
22
23 for (int x = maxc; x >= 1; x--) {
24     ll cnt_mul = 0; // number of multiples of x
25     for (int y = x; y <= maxc; y += x)
26         cnt_mul += cnt[y];
27
28     dp[x] = cnt_mul * (cnt_mul - 1) / 2; // number
   of pairs that are divisible by x
29     for (int y = x + x; y <= maxc; y += x)
30         dp[x] -= dp[y]; // PIE: subtract all dp[y]
   for y > x and x|y
31 }

```

10.3 Count Number of Divisors

```

1 // Author: Gino
2 // Function: Count the number of divisors for all x
   <= 10^6 using harmonic series
3 const int maxc = 1e6;
4 vector<int> facs;
5
6 void find_all_divisors() {
7     facs.clear(); facs.resize(maxc + 1, 0);
8     for (int x = 1; x <= maxc; x++) {
9         for (int y = x; y <= maxc; y += x) {
10             facs[y]++;
11         }
12     }
13 }

```

10.4 數論分塊

```

1 // Author: Gino
2 /*
3 n = 17
4 i:  1  2  3  4  5  6  7  8  9 10 11 12 13 14 15
   16 17
5 n/i: 17  8  5  4  3  2  2  2  1  1  1  1  1  1  1
   1  1
6
7             ^      ^
8             L(2)  R(2)
9
9 L(x) := left bound for n/i = x
10 R(x) := right bound for n/i = x
11
12 ===== FORMULA =====
13 >>> R = n / (n/L) <<<
14 =====
15
16 Example: L(2) = 6

```

```

17         R(2) = 17 / (17 / 6)
18             = 17 / 2
19             = 8
20 */
21 // ===== CODE =====
22
23 for (ll l = 1, r = 1, q = n; l <= n; l = r + 1) {
24     q = n/l;
25     r = n/q;
26     // Process your code here
27 }
28 // q, l, r: 17 1 1
29 // q, l, r: 8 2 2
30 // q, l, r: 5 3 3
31 // q, l, r: 4 4 4
32 // q, l, r: 3 5 5
33 // q, l, r: 2 6 8
34 // q, l, r: 1 9 17

```

10.5 Pollard's rho

```

1 // Author: Unknown
2 // Function: Find a non-trivial factor of a big
   number in  $O(n^{1/4} \log^2(n))$ 
3
4 ll find_factor(ll number) {
5     __int128 x = 2;
6     for (__int128 cycle = 1; ; cycle++) {
7         __int128 y = x;
8         for (int i = 0; i < (1<<cycle); i++) {
9             x = (x * x + 1) % number;
10            __int128 factor = __gcd(x - y, number);
11            if (factor > 1)
12                return factor;
13        }
14    }
15 }

```

```

1 # Author: Unknown
2 # Function: Find a non-trivial factor of a big
   number in  $O(n^{1/4} \log^2(n))$ 
3 from itertools import count
4 from math import gcd
5 from sys import stdin
6
7 for s in stdin:
8     number, x = int(s), 2
9     brk = False
10    for cycle in count(1):
11        y = x
12        if brk:
13            break
14        for i in range(1<< cycle):
15            x = (x * x + 1) % number
16            factor = gcd(x - y, number)
17            if factor > 1:
18                print(factor)
19                brk = True
20                break

```

10.6 Miller Rabin

```

1 // Author: Unknown, Modified by Gino
2 // Function: Check if a number is a prime in  $O(100
   * \log^2(n))$ 
3 //     miller_rabin(): return 1 if prime, 0
   otherwise
4 inline ll mul(ll x, ll y, ll mod) {

```



```

5     return (__int128)(x) * y % mod;
6 }
7 ll mypow(ll a, ll b, ll mod) {
8     ll r = 1;
9     while (b > 0) {
10         if (b & 1) r = mul(r, a, mod);
11         a = mul(a, a, mod);
12         b >>= 1;
13     }
14     return r;
15 }
16 bool witness(ll a, ll n, ll u, int t){
17     ll x = mypow(a, u, n);
18     for(int i = 0; i < t; i++) {
19         ll nx = mul(x, x, n);
20         if (nx == 1 && x != 1 && x != n-1) return true;
21         x = nx;
22     }
23     return x != 1;
24 }
25 bool miller_rabin(ll n) {
26     // if n >= 3,474,749,660,383
27     // change {2, 3, 5, 7, 11, 13} to
28     // {2, 325, 9375, 28178, 450775, 9780504,
29     // 1795265022}
30     if (n < 2) return false;
31     if(!(n & 1)) return n == 2;
32     ll u = n - 1; int t = 0;
33     while(!(u & 1)) u >>= 1, t++;
34     for (ll a : {2, 3, 5, 7, 11, 13}) {
35         if (a % n == 0) continue;
36         if (witness(a, n, u, t)) return false;
37     }
38     return true;
39 }
40 // bases that make sure no pseudoprimes flee from
41 // test
42 // if WA, replace randll(n - 1) with these bases:
43 // n < 4,759,123,141      3 : 2, 7, 61
44 // n < 1,122,004,669,633 4 : 2, 13, 23, 1662803
45 // n < 3,474,749,660,383 6 : pirmses <= 13
46 // n < 2^64              7 :
47 // 2, 325, 9375, 28178, 450775, 9780504, 1795265022

```

10.7 Discrete Log

```

1 // Author: ckiseki
2 // find  $x^? = y \pmod{M}$ 
3 //  $O(\sqrt{M})$ 
4 template<typename Int>
5 Int BSGS(Int x, Int y, Int M) {
6     //  $x^? \equiv y \pmod{M}$ 
7     Int t = 1, c = 0, g = 1;
8     for (Int M_ = M; M_ > 0; M_ >>= 1) g = g * x % M;
9     for (g = gcd(g, M); t % g != 0; ++c) {
10         if (t == y) return c;
11         t = t * x % M;
12     }
13     if (y % g != 0) return -1;
14     t /= g, y /= g, M /= g;
15     Int h = 0, gs = 1;
16     for (; h * h < M; ++h) gs = gs * x % M;
17     unordered_map<Int, Int> bs;
18     for (Int s = 0; s < h; bs[y] = ++s) y = y * x %
19     M;
20     for (Int s = 0; s < M; s += h) {

```

```

20     t = t * gs % M;
21     if (bs.count(t)) return c + s + h - bs[t];
22 }
23 return -1;
24 }

```

10.8 Discrete Sqrt

```

1 // Author: ckiseki
2 // find solution for  $x^2 = n \pmod{P}$ 
3 //  $O(\log^2 P)$ 
4 int get_root(int n, int P) { // ensure  $0 \leq n < P$ 
5     if (P == 2 or n == 0) return n;
6     auto check = [&](lld x) {
7         return modpow(int(x), (P - 1) / 2, P); };
8     if (check(n) != 1) return -1;
9     mt19937 rnd(7122); lld z = 1, w;
10    while (check(w = (z * z - n + P) % P) != P - 1)
11        z = rnd() % P;
12    const auto M = [P, w](auto &u, auto &v) {
13        auto [a, b] = u; auto [c, d] = v;
14        return make_pair((a * c + b * d % P * w) % P,
15            (a * d + b * c) % P);
16    };
17    pair<lld, lld> r(1, 0), e(z, 1);
18    for (int q = (P + 1) / 2; q; q >>= 1, e = M(e,
19        e))
20        if (q & 1) r = M(r, e);
21    return int(r.first); // sqrt(n) mod P where P is
    prime

```

10.9 Fast Power

Note: $a^n \equiv a^{(n \bmod (p-1))} \pmod{p}$

10.10 Extend GCD

```

1 // Author: Gino
2 // [Usage]
3 // bezout(a, b, c):
4 //     find solution to  $ax + by = c$ 
5 //     return {-LINF, -LINF} if no solution
6 // inv(a, p):
7 //     find modulo inverse of a under p
8 //     return -1 if not exist
9 // CRT(vector<ll>& a, vector<ll>& m)
10 //     find a solution pair (x, mod) satisfies all
11 //      $x = a[i] \pmod{m[i]}$ 
12 //     return {-LINF, -LINF} if no solution
13 const ll LINF = 4e18;
14 typedef pair<ll, ll> pll;
15 template<typename T1, typename T2>
16 T1 chmod(T1 a, T2 m) {
17     return (a % m + m) % m;
18 }
19
20 ll GCD;
21 pll extgcd(ll a, ll b) {
22     if (b == 0) {
23         GCD = a;
24         return pll{1, 0};
25     }
26     pll ans = extgcd(b, a % b);
27     return pll{ans.second, ans.first - a/b *
28         ans.second};
29 }

```

```

29 pll bezout(ll a, ll b, ll c) {
30     bool negx = (a < 0), negy = (b < 0);
31     pll ans = extgcd(abs(a), abs(b));
32     if (c % GCD != 0) return pll{-LINF, -LINF};
33     return pll{ans.first * c/GCD * (negx ? -1 : 1),
34               ans.second * c/GCD * (negy ? -1 :
35 1)};
36 }
37 ll inv(ll a, ll p) {
38     if (p == 1) return -1;
39     pll ans = bezout(a % p, -p, 1);
40     if (ans == pll{-LINF, -LINF}) return -1;
41     return chmod(ans.first, p);
42 }
43 pll CRT(vector<ll>& a, vector<ll>& m) {
44     for (int i = 0; i < (int)a.size(); i++)
45         a[i] = chmod(a[i], m[i]);
46
47     ll x = a[0], mod = m[0];
48     for (int i = 1; i < (int)a.size(); i++) {
49         pll sol = bezout(mod, m[i], a[i] - x);
50         if (sol.first == -LINF) return pll{-LINF, -
51 LINF};
52
53         // prevent long long overflow
54         ll p = chmod(sol.first, m[i] / GCD);
55         ll lcm = mod / GCD * m[i];
56         x = chmod((__int128)p * mod + x, lcm);
57         mod = lcm;
58     }
59     return pll{x, mod};
60 }

```

10.11 Mu + Phi

```

1 // Author: Gino
2 const int maxn = 1e6 + 5;
3 ll f[maxn];
4 vector<int> lpf, prime;
5 void build() {
6     lpf.clear(); lpf.resize(maxn, 1);
7     prime.clear();
8     f[1] = ...; /* mu[1] = 1, phi[1] = 1 */
9     for (int i = 2; i < maxn; i++) {
10         if (lpf[i] == 1) {
11             lpf[i] = i; prime.emplace_back(i);
12             f[i] = ...; /* mu[i] = 1, phi[i] = i-1 */
13         }
14         for (auto& j : prime) {
15             if (i*j >= maxn) break;
16             lpf[i*j] = j;
17             if (i % j == 0) f[i*j] = ...; /* 0,
18 phi[i]*j */
19             else f[i*j] = ...; /* -mu[i],
20 phi[i]*phi[j] */
21             if (j >= lpf[i]) break;
22         }
23     }
24 }

```

10.12 Other Formulas

- Pisano Period: 任何線性遞迴 (比如費氏數列) 模任何一個數字 M 都會循環, 找循環節 $\pi(M)$ 先質因數分解 $M = \prod p_i^{e_i}$, 然後 $\pi(M) = \text{lcm}(\pi(p_i^{e_i}))$,
- Inversion: $aa^{-1} \equiv 1 \pmod{m}$. a^{-1} exists iff $\gcd(a, m) = 1$.
- Linear inversion: $a^{-1} \equiv (m - \lfloor \frac{m}{a} \rfloor) \times (m \bmod a)^{-1} \pmod{m}$

- Fermat's little theorem: $a^p \equiv a \pmod{p}$ if p is prime.
- Euler function: $\varphi(n) = n \prod_{p|n} \frac{p-1}{p}$
- Euler theorem: $a^{\varphi(n)} \equiv 1 \pmod{n}$ if $\gcd(a, n) = 1$. If a, n are not coprime: 質因數分解 $n = \prod p_i^{e_i}$, 對每個 $p_i^{e_i}$ 分開看他們跟 a 是否互質 (互質: Fermat / 不互質: 夠大的指數會直接削成 0), 最後用 CRT 合併.
- Extended Euclidean algorithm: $ax + by = \gcd(a, b) = \gcd(b, a \bmod b) = \gcd(b, a - \lfloor \frac{a}{b} \rfloor b) = bx_1 + (a - \lfloor \frac{a}{b} \rfloor b)y_1 = ay_1 + b(x_1 - \lfloor \frac{a}{b} \rfloor y_1)$
- Divisor function: $\sigma_x(n) = \sum_{d|n} d^x \cdot n = \prod_{i=1}^r p_i^{a_i}$.
 $\sigma_x(n) = \prod_{i=1}^r \frac{p_i^{(a_i+1)x} - 1}{p_i^x - 1}$ if $x \neq 0$. $\sigma_0(n) = \prod_{i=1}^r (a_i + 1)$.
- Chinese remainder theorem (Coprime Moduli):
 $x \equiv a_i \pmod{m_i}$.
 $M = \prod m_i$. $M_i = \frac{M}{m_i}$. $t_i = M_i^{-1}$.
 $x = kM + \sum a_i t_i M_i$, $k \in \mathbb{Z}$.
- Chinese remainder theorem:
 $x \equiv a_1 \pmod{m_1}, x \equiv a_2 \pmod{m_2} \Rightarrow x = m_1 p + a_1 = m_2 q + a_2 \Rightarrow m_1 p - m_2 q = a_2 - a_1$
Solve for (p, q) using ExtGCD.
 $x \equiv m_1 p + a_1 \equiv m_2 q + a_2 \pmod{\text{lcm}(m_1, m_2)}$
- Avoiding Overflow: $ca \bmod cb = c(a \bmod b)$
- Dirichlet Convolution: $(f * g)(n) = \sum_{d|n} f(d)g(\frac{n}{d})$
- Important Multiplicative Functions + Properties:
 - $\varepsilon(n) = [n = 1]$
 - $1(n) = 1$
 - $id(n) = n$
 - $\mu(n) = 0$ if n has squared prime factor
 - $\mu(n) = (-1)^k$ if $n = p_1 p_2 \cdots p_k$
 - $\varepsilon = \mu * 1$
 - $\varphi = \mu * id$
 - $[n = 1] = \sum_{d|n} \mu(d)$
 - $[\gcd = 1] = \sum_{d|\gcd} \mu(d)$
- Möbius inversion: $f = g * 1 \Leftrightarrow g = f * \mu$

10.13 Polynomial

```

1 // Author: Gino
2 // Preparation: first set_mod(mod, g), then
3 // init_ntt()
4 // everytime you change the mod, you have to call
5 // init_ntt() again
6 //
7 // [Usage]
8 // polynomial: vector<ll> a, b
9 // negation: -a
10 // add/subtract: a += b, a -= b
11 // convolution: a *= b
12 // in-place modulo: mod(a, b)
13 // in-place inversion under mod x^N: inv(ia, N)
14
15 const int maxk = 20;
16 const int maxn = 1<<maxk;
17
18 using u64 = unsigned long long;
19 using u128 = __uint128_t;
20
21 int g;
22 u64 MOD;
23 u64 BARRETT_IM; // [ 2^64 / MOD ]
24
25 inline void set_mod(u64 m, int _g) {
26     g = _g;
27 }

```

```

26 MOD = m;
27 BARRETT_IM = (u128(1) << 64) / m;
28 }
29 inline u64 chmod(u128 x) {
30     u64 q = (u64)((x * BARRETT_IM) >> 64);
31     u64 r = (u64)(x - (u128)q * MOD);
32     if (r >= MOD) r -= MOD;
33     return r;
34 }
35 inline u64 mmul(u64 a, u64 b) {
36     return chmod((u128)a * b);
37 }
38 ll pw(ll a, ll n) {
39     ll ret = 1;
40     while (n > 0) {
41         if (n & 1) ret = mmul(ret, a);
42         a = mmul(a, a);
43         n >>= 1;
44     }
45     return ret;
46 }
47
48 vector<ll> X, iX;
49 vector<int> rev;
50 void init_ntt() {
51     X.assign(maxn, 1); //  $x_1 = g^{((p-1)/n)}$ 
52     iX.assign(maxn, 1);
53
54     ll u = pw(g, (MOD-1)/maxn);
55     ll iu = pw(u, MOD-2);
56     for (int i = 1; i < maxn; i++) {
57         X[i] = mmul(X[i-1], u);
58         iX[i] = mmul(iX[i-1], iu);
59     }
60
61     if ((int)rev.size() == maxn) return;
62     rev.assign(maxn, 0);
63     for (int i = 1, hb = -1; i < maxn; i++) {
64         if (!(i & (i-1))) hb++;
65         rev[i] = rev[i ^ (1<<hb)] | (1<<(maxk-
66 hb-1));
67     }
68
69     template<typename T>
70     void NTT(vector<T>& a, bool inv=false) {
71         int _n = (int)a.size();
72         int k = __lg(_n) + ((1<<__lg(_n)) != _n);
73         int n = 1<<k;
74         a.resize(n, 0);
75
76         short shift = maxk-k;
77         for (int i = 0; i < n; i++)
78             if (i > (rev[i]>>shift))
79                 swap(a[i], a[rev[i]>>shift]);
80         for (int len = 2, half = 1, div = maxn>>1; len
81 <= n; len<<=1, half<<=1, div>>=1) {
82             for (int i = 0; i < n; i += len) {
83                 for (int j = 0; j < half; j++) {
84                     T u = a[i+j];
85                     T v = mmul(a[i+j+half], (inv ?
86 iX[j*div] : X[j*div]));
87                     a[i+j] = (u+v >= MOD ? u+v-MOD :
88 u+v);
89                     a[i+j+half] = (u-v < 0 ? u-v+MOD :
90 u-v);
91                 }
92             }
93         }
94     }
95 }

```

```

86 if (inv) {
87     T dn = pw(n, MOD-2);
88     for (auto& x : a) {
89         x = mmul(x, dn);
90     }
91 }
92
93 template<typename T>
94 inline void shrink(vector<T>& a) {
95     int cnt = (int)a.size();
96     for (; cnt > 0; cnt--) if (a[cnt-1]) break;
97     a.resize(max(cnt, 1));
98 }
99
100 template<typename T>
101 vector<T>& operator*=(vector<T>& a, vector<T> b) {
102     int na = (int)a.size();
103     int nb = (int)b.size();
104     a.resize(na + nb - 1, 0);
105     b.resize(na + nb - 1, 0);
106
107     NTT(a); NTT(b);
108     for (int i = 0; i < (int)a.size(); i++)
109         a[i] = mmul(a[i], b[i]);
110     NTT(a, true);
111
112     shrink(a);
113     return a;
114 }
115
116 inline ll crt(ll a0, ll a1, ll m1, ll m2, ll
117 inv_m1_mod_m2){
118     //  $x \equiv a_0 \pmod{m_1}$ ,  $x \equiv a_1 \pmod{m_2}$ 
119     //  $t = (a_1 - a_0) * \text{inv}(m_1) \pmod{m_2}$ 
120     //  $x = a_0 + t * m_1 \pmod{m_1 * m_2}$ 
121     ll t = chmod(a1 - a0);
122     if (t < 0) t += m2;
123     t = (ll)((__int128)t * inv_m1_mod_m2 % m2);
124     return a0 + (ll)((__int128)t * m1);
125 }
126
127 void mul_crt() {
128     // a copy to a1, a2 | b copy to b1, b2
129     ll M1 = 998244353, M2 = 1004535809;
130     g = 3; set_mod(M1); init_ntt(); a1 *= b1;
131     g = 3; set_mod(M2); init_ntt(); a2 *= b2;
132
133     ll inv_m1_mod_m2 = pw(M1, M2 - 2);
134     for (int i = 2; i <= 2 * k; i++)
135         cout << crt(a1[i], a2[i], M1, M2,
136 inv_m1_mod_m2) << ' ';
137     cout << endl;
138 }
139
140 /*  $P = r * 2^k + 1$ 
141
142 P          r    k    g
143 998244353  119  23   3
144 1004535809 479  21   3
145
146 P          r    k    g
147 3          1    1    2
148 5          1    2    2
149 17         1    4    3
150 97         3    5    5
151 193        3    6    5
152 257        1    8    3
153 7681       15    9   17
154 12289      3   12   11
155 40961      5   13    3

```

```

148 65537          1  16  3
149 786433        3  18 10
150 5767169       11  19  3
151 7340033       7  20  3
152 23068673      11  21  3
153 104857601     25  22  3
154 167772161     5  25  3
155 469762049     7  26  3
156 1004535809   479  21  3
157 2013265921   15  27 31
158 2281701377   17  27  3
159 3221225473    3  30  5
160 75161927681  35  31  3
161 77309411329   9  33  7
162 206158430209  3  36 22
163 2061584302081 15  37  7
164 2748779069441 5  39  3
165 6597069766657 3  41  5
166 3958241859937 9  42  5
167 79164837199873 9  43  5
168 263882790666241 15 44  7
169 1231453023109121 35 45  3
170 1337006139375617 19 46  3
171 3799912185593857 27 47  5
172 4222124650659841 15 48 19
173 7881299347898369 7  50  6
174 31525197391593473 7  52  3
175 180143985094819841 5  55  6
176 1945555039024054273 27 56  5
177 4179340454199820289 29 57  3
178 9097271247288401921 505 54  6 */

```

10.14 Counting Primes

```

1 // prime_count - #primes in [1..n] ( $O(n^{2/3})$ 
  time,  $O(\sqrt{n})$  memory)
2
3 using u64 = unsigned long long;
4 static inline u64 prime_count(u64 n){
5     if(n<=1) return 0;
6     int v = (int)floor(sqrt((long double)n));
7     int s = (v+1)>>1, pc = 0;
8     vector<int> smalls(s), roughs(s), skip(v+1);
9     vector<long long> larges(s);
10
11     for(int i=0; i<s; ++i){
12         smalls[i]=i;
13         roughs[i]=2*i+1;
14         larges[i]=(long long)((n/roughs[i]-1)>>1);
15     }
16
17     for(int p=3; p<=v; p+=2) if(!skip[p]){
18         int q = p*p;
19         if(1LL*q*q > (long long)n) break;
20         skip[p]=1;
21         for(int i=q; i<=v; i+=2*p) skip[i]=1;
22
23         int ns=0;
24         for(int k=0; k<s; ++k){
25             int i = roughs[k];
26             if(skip[i]) continue;
27             u64 d = (u64)i * (u64)p;
28             long long sub = (d <= (u64)v)
29                 ? larges[smalls[(int)(d>>1)] - pc]
30                 : smalls[(int)((n/d - 1) >> 1)];
31             larges[ns] = larges[k] - sub + pc;

```

```

32         roughs[ns++] = i;
33     }
34     s = ns;
35     for(int i=(v-1)>>1, j=((v/p)-1)|1; j>=p; j-=2){
36         int c = smalls[j>>1] - pc;
37         for(int e=(j*p)>>1; i>=e; --i) smalls[i] -=
38             c;
39         ++pc;
40     }
41
42     larges[0] += 1LL*(s + 2*(pc-1))*(s-1) >> 1;
43     for(int k=1; k<s; ++k) larges[0] -= larges[k];
44
45     for(int l=1; l<s; ++l){
46         int q = roughs[l];
47         u64 m = n / (u64)q;
48         long long t = 0;
49         int e = smalls[(int)((m/q - 1) >> 1)] - pc;
50         if(e < l+1) break;
51         for(int k=l+1; k<=s; ++k) t += smalls[(int)((m/
52             (u64)roughs[k] - 1) >> 1)];
53         larges[0] += t - 1LL*(e - l)*(pc + l - 1);
54     }
55     return (u64)(larges[0] + 1);
56 }

```

10.15 Linear Sieve for Other Number Theoretic Functions

```

1 // linear_sieve(n, primes, lp, phi, mu, d, sigma)
2 // Outputs over the index range 0..n (n >= 1):
3 //   primes : all primes in [2..n], increasing.
4 //   lp      : lowest prime factor; lp[1]=0, lp[x]
  is the smallest prime dividing x.
5 //   phi     : Euler totient, phi[x] =  $|\{1 \leq k \leq x : \gcd(k, x)=1\}|$ . Multiplicative.
6 //   mu      : Möbius; mu[1]=1, mu[x]=0 if x has a
  squared prime factor, else  $(-1)^{\#\text{distinct primes}}$ .
7 //   d       : number of divisors; if  $x = \prod p_i^{e_i}$ ,
  then  $d[x] = \prod (e_i + 1)$ . Multiplicative.
8 //   sigma   : sum of divisors; if  $x = \prod p_i^{e_i}$ ,
  then  $\sigma[x] = \prod (1 + p_i + \dots + p_i^{e_i})$ . (use ll)
9 //
10 // Complexity:  $O(n)$  time,  $O(n)$  memory.
11 // Notes: Arrays are resized inside; primes is
  cleared and reserved. sigma uses ll to avoid 32-bit
  overflow.
12
13 static inline void linear_sieve(
14     int n,
15     std::vector<int> &primes,
16     std::vector<int> &lp,
17     std::vector<int> &phi,
18     std::vector<int> &mu,
19     std::vector<int> &d,
20     std::vector<ll> &sigma
21 ) {
22     lp.assign(n + 1, 0); phi.assign(n + 1, 0);
23     mu.assign(n + 1, 0); d.assign(n + 1, 0);
24     sigma.assign(n + 1, 0);
25     primes.clear(); primes.reserve(n > 1 ? n / 10 :
26         0);
27     std::vector<int> cnt(n + 1, 0), core(n + 1, 1);
28     std::vector<ll> p_pow(n + 1, 1), sum_p(n + 1,
29         1);
30     phi[1] = mu[1] = d[1] = sigma[1] = 1;
31
32     for (int i = 2; i <= n; ++i) {

```

```

29     if (!lp[i]) {
30         lp[i] = i; primes.push_back(i);
31         phi[i] = i - 1; mu[i] = -1; d[i] = 2;
32         cnt[i] = 1; p_pow[i] = i; core[i] = 1;
33         sum_p[i] = 1 + (ll)i; sigma[i] = sum_p[i];
34     }
35     for (int p : primes) {
36         long long ip = 1LL * i * p;
37         if (ip > n) break;
38         lp[ip] = p;
39         if (p == lp[i]) {
40             cnt[ip] = cnt[i] + 1; p_pow[ip] = p_pow[i]
41             * p; core[ip] = core[i];
42             sum_p[ip] = sum_p[i] + p_pow[ip];
43             phi[ip] = phi[i] * p; mu[ip] = 0;
44             d[ip] = d[core[ip]] * (cnt[ip] + 1);
45             sigma[ip] = sigma[core[ip]] * sum_p[ip];
46             break; // critical for linear complexity
47         } else {
48             cnt[ip] = 1; p_pow[ip] = p; core[ip] = i;
49             sum_p[ip] = 1 + (ll)p;
50             phi[ip] = phi[i] * (p - 1); mu[ip] = -
51             mu[i];
52             d[ip] = d[i] * 2;
53             sigma[ip] = sigma[i] * sum_p[ip];
54         }
55     }
56
57 // Optional helper: factorize x in O(log x) using
58 // lp (requires x in [2..n])
59 static inline std::vector<std::pair<int,int>>
60 factorize(int x, const std::vector<int>& lp) {
61     std::vector<std::pair<int,int>> res;
62     while (x > 1) {
63         int p = lp[x], e = 0;
64         do { x /= p; ++e; } while (x % p == 0);
65         res.push_back({p, e});
66     }
67     return res;
68 }

```

11 Linear Algebra

11.1 Gaussian-Jordan Elimination

```

1 int n; vector<vector<ll>> v;
2 void gauss(vector<vector<ll>>& v) {
3     int r = 0;
4     for (int i = 0; i < n; i++) {
5         bool ok = false;
6         for (int j = r; j < n; j++) {
7             if (v[j][i] == 0) continue;
8             swap(v[j], v[r]);
9             ok = true; break;
10        }
11        if (!ok) continue;
12        ll div = inv(v[r][i]);
13        for (int j = 0; j < n+1; j++) {
14            v[r][j] *= div;
15            if (v[r][j] >= MOD) v[r][j] %= MOD;
16        }
17        for (int j = 0; j < n; j++) {
18            if (j == r) continue;
19            ll t = v[j][i];

```

```

20         for (int k = 0; k < n+1; k++) {
21             v[j][k] -= v[r][k] * t % MOD;
22             if (v[j][k] < 0) v[j][k] += MOD;
23         }
24         r++;
25     } }

```

11.2 Determinant

1. Use GJ Elimination, if there's any row consists of only 0, then $\det = 0$, otherwise $\det = \text{product of diagonal elements}$.
2. Properties of $\det$:
 - Transpose: Unchanged
 - Row Operation 1 - Swap 2 rows: $-\det$
 - Row Operation 2 - $k\vec{r}_i$: $k \times \det$
 - Row Operation 3 - $k\vec{r}_i$ add to $\vec{r}_j$: Unchanged

12 Flow / Matching

12.1 Flow Methods

- 1 // Author: CRYPTOGRAPHER (some modified by Gino)
- 2 Maximize $c^T x$ subject to $Ax \leq b, x \geq 0$;
- 3 with the corresponding symmetric dual problem,
- 4 Minimize $b^T y$ subject to $A^T y \geq c, y \geq 0$.
- 5
- 6 Maximize $c^T x$ subject to $Ax \leq b$;
- 7 with the corresponding asymmetric dual problem,
- 8 Minimize $b^T y$ subject to $A^T y = c, y \geq 0$.
- 9
- 10 Maximize $\sum x$ subject to $x_i + x_j \leq A_{ij}, x \geq 0$;
- 11 $\Rightarrow$ Maximize $\sum x$ subject to $x_i + x_j \leq A_{ij}$;
- 12 $\Rightarrow$ Minimize $A^T y = \sum A_{ij} y_{ij}$ subject to for
- 13 all v , $\sum_{i=v \text{ or } j=v} y_{ij} = 1, y_{ij} \geq 0$
- 14 $\Rightarrow$ possible optimal solution: $y_{ij} = \{0, 0.5, 1\}$
- 15 $\Rightarrow y' = 2y$: $\sum_{i=v \text{ or } j=v} y'_{ij} = 2, y'_{ij} = \{0, 1, 2\}$
- 16
- 17 $\Rightarrow$ Minimum Bipartite perfect matching/2
- 18 ($V1=X, V2=X, E=A$)
- 19
- 20 General Graph:
- 21 $|\text{Max Ind. Set}| + |\text{Min Vertex Cover}| = |V|$
- 22 $|\text{Max Ind. Edge Set}| + |\text{Min Edge Cover}| = |V|$
- 23
- 24 Bipartite Graph:
- 25 $|\text{Max Ind. Set}| = |\text{Min Edge Cover}|$
- 26 $|\text{Max Ind. Edge Set}| = |\text{Min Vertex Cover}|$
- 27
- 28 To reconstruct the minimum vertex cover, dfs from
- 29 each
- 30 unmatched vertex on the left side and with unused
- 31 edges
- 32 only. Equivalently, dfs from source with unused
- 33 edges
- 34 only and without visiting sink. Then, a vertex is
- 35 chosen iff. it is on the left side and without
- 36 visited
- 37 or on the right side and visited through dfs.
- 38
- 39 Minimum Weighted Bipartite Edge Cover:
- 40 Construct new bipartite graph with $n+m$ vertices on
- 41 each side:
- 42 for each vertex u , duplicate a vertex u' on the
- 43 other side
- 44 for each edge (u,v,w) , add edges (u,v,w) and
- 45 (v',u',w)
- 46 for each vertex u , add edge $(u,u',2w)$ where w is
- 47 min edge connects to u
- 48 then the answer is the minimum perfect matching of
- 49 the new graph (KM)

```

37
38 Maximum density subgraph (  $\sum W_e + \sum W_v$  ) / |V|
39 Binary search on answer:
40 For a fixed D, construct a Max flow model as follow:
41 Let S be Sum of all weight( or inf)
42 1. from source to each node with cap = S
43 2. For each (u,v,w) in E, (u->v, cap=w), (v->u, cap=w)
44 3. For each node v, from v to sink with cap = S + 2 * D - deg[v] - 2 * (W of v)
45 where deg[v] =  $\sum$  weight of edge associated with v
46 If maxflow < S * |V|, D is an answer.
47
48 Requiring subgraph: all vertex can be reached from source with
49 edge whose cap > 0.
50
51 Maximum closed subgraph
52 1. connect source with positive weighted vertex(capacity=weight)
53 2. connect sink with negative weighted vertex(capacity=-weight)
54 3. make capacity of the original edges = inf
55 4. ans = sum(positive weighted vertex weight) - (max flow)
56
57 Node-disjoint) Min DAG Path Cover (用最少路徑覆蓋所有點)
58 Node disjoint: 拆出來的路徑不能共用同一個點
59 將一個點 u 裂成 u+ 和 u-, 代表進入和出去
60 對於一條邊 (u, v) 建邊 u- => v+
61 1. +點們和 -點們形成一張二分圖
62 2. 最差的答案是 n, 代表每個點自己一個點就是一條路徑
63 3. 只要一組 (u-, v+) 匹配成功那對應到的答案恰好會 -1
64 >>> ans = n - 這張二分圖的最大匹配
65
66 General DAG Path Cover
67 跟 Node-disjoint 版本差在點可以被很多條路共用
68 建邊方式 Node-disjoint 差別在條件比較鬆
69 Node-disjoint: u- => v+ 只能在 (u, v) 有邊時建
70 General: u- => v+ 在 u 能走到 v 的時候就建
71
72 Dilworth Theorem
73 反鏈: 一些節點的集合, 滿足這些節點互相無法抵達
74 最大反鏈 = 最小 General DAG Path Cover

```

12.2 Dinic

```

1 // Author: Benson (Extensions by Gino)
2 // Function: Max Flow,  $O(V^2 E)$ 
3 // Usage: Call init(n) first, then add(u, v, w) based on your model
4 // (!) vertices 0-based
5 // >>> flow() := return max flow
6 // >>> find_cut() := return min cut + store cut set in dinic.cut
7 // >>> find_matching() := return |M| + store matching plan in dinic.matching
8 // >>> flow_decomposition := return max flow + store decomposition in dinic.D
9 #define int long long
10 #define eb emplace_back
11 #define ALL(a) a.begin(), a.end()
12 struct Dinic {
13     struct Edge {
14         // t: to | C: original capacity | c: current capacity | r: residual edge | f: current flow

```

```

15     int t, C, c, r, f;
16     bool fw; // is in forward-edge graph
17     Edge() {}
18     Edge(int _t, int _C, int _r, bool _fw, int _f=0):
19         t(_t), C(_C), c(_C), r(_r), fw(_fw), f(_f) {}
20 };
21 vector<vector<Edge>> G;
22 vector<int> dis, iter;
23 int n, s, t;
24 void init(int _n) {
25     n = _n;
26     G.resize(n), dis.resize(n), iter.resize(n);
27     for(int i = 0; i < n; ++i)
28         G[i].clear();
29 }
30 void add(int a, int b, int c) {
31     G[a].eb(b, c, G[b].size(), true);
32     G[b].eb(a, 0, G[a].size() - 1, false);
33 }
34 bool bfs() {
35     fill(ALL(dis), -1);
36     dis[s] = 0;
37     queue<int> que;
38     que.push(s);
39     while(!que.empty()) {
40         int u = que.front(); que.pop();
41         for(auto& e : G[u]) {
42             if(e.c > 0 && dis[e.t] == -1) {
43                 dis[e.t] = dis[u] + 1;
44                 que.push(e.t);
45             }
46         }
47     }
48     return dis[t] != -1;
49 }
50 int dfs(int u, int cur) {
51     if(u == t) return cur;
52     for(int &i = iter[u]; i < (int)G[u].size(); ++i) {
53         auto& e = G[u][i];
54         if(e.c > 0 && dis[u] + 1 == dis[e.t]) {
55             int ans = dfs(e.t, min(cur, e.c));
56             if(ans > 0) {
57                 G[e.t][e.r].c += ans;
58                 e.c -= ans;
59                 return ans;
60             }
61         }
62     }
63     return 0;
64 }
65 // find max flow
66 int flow(int a, int b) {
67     s = a, t = b;
68     int ans = 0;
69     while(bfs()) {
70         fill(ALL(iter), 0);
71         int tmp;
72         while((tmp = dfs(s, INF)) > 0)
73             ans += tmp;
74     }
75     return ans;
76 }
77 // min cut plan
78 vector<pair<pair<int, int>, int>> cut;
79 int find_cut(int a, int b) {
80     int cut_sz = flow(a, b);

```



```

77 vector<int> vis(n, 0);
78 cut.clear();
79 function<void(int)> dfs = [&](int u) {
80     vis[u] = 1;
81     for (auto& e : G[u])
82         if (e.c > 0 && !vis[e.t])
83             dfs(e.t);
84 };
85 dfs(a);
86 for (int u = 0; u < n; u++)
87     if (vis[u])
88         for (auto& e : G[u])
89             if (!vis[e.t])
90                 cut.eb(make_pair(make_pair(u, e.t),
91 G[e.t][e.r].c));
92 return cut.sz;
93 }
94 // bipartite matching plan
95 vector<pair<int, int>> matching;
96 int find_matching(int Xstart, int Xend, int
97 Ystart, int Yend, int a, int b) {
98     int msz = flow(a, b);
99     matching.clear();
100     for (int x = Xstart; x <= Xend; x++)
101         for (auto& e : G[x])
102             if (e.c == 0 && Ystart <= e.t && e.t <=
103 Yend)
104                 matching.emplace_back(make_pair(x, e.t));
105 return msz;
106 }
107 // flow decomposition
108 vector<pair<int, vector<int>>> D; // (flow
109 amount, [path p1 ... pk])
110 int flow_decomposition(int a, int b) {
111     int mxflow = flow(a, b);
112     vector<vector<Edge>> fG(n); // graph consists
113 of forward edges
114 for (int u = 0; u < n; u++) {
115     for (auto& e : G[u]) {
116         if (e.fw) {
117             e.f = e.c - e.c;
118             if (e.f > 0) fG[u].eb(e);
119         } } }
120 vector<int> vis;
121 function<int(int, int)> dfs = [&](int u, int
122 cur) {
123     if (u == b) {
124         D.back().second.eb(u);
125         return cur;
126     }
127     vis[u] = 1;
128     for (auto& e : fG[u]) {
129         if (e.f > 0 && !vis[e.t]) {
130             int ans = dfs(e.t, min(cur, e.f));
131             if (ans > 0) {
132                 e.f -= ans;
133                 D.back().second.eb(u);
134                 return ans;
135             } } }
136     return 0LL;
137 };
138 D.clear();
139 int quota = mxflow;

```

```

136 while (quota > 0) {
137     D.emplace_back(make_pair(0, vector<int>()));
138     vis.assign(n, 0);
139     int f = dfs(a, INF);
140     if (f == 0) break;
141     reverse(D.back().second.begin(),
142 D.back().second.end());
143     D.back().first = f, quota -= f;
144 }
145 return mxflow;
146 };

```

12.3 ISAP

```

1 // Author: CRYPTOGRAPHER
2 #define SZ(c) ((int)(c).size())
3 static const int MAXV=50010;
4 static const int INF=1000000;
5 struct Maxflow{
6     struct Edge{
7         int v,c,r;
8         Edge(int _v,int _c,int _r):v(_v),c(_c),r(_r){}
9     };
10     int s,t; vector<Edge> G[MAXV];
11     int iter[MAXV],d[MAXV],gap[MAXV],tot;
12     void init(int n,int _s,int _t){
13         tot=n,s=_s,t=_t;
14         for(int i=0;i<=tot;i++){
15             G[i].clear(); iter[i]=d[i]=gap[i]=0;
16         }
17     }
18     void addEdge(int u,int v,int c){
19         G[u].push_back(Edge(v,c,SZ(G[v])));
20         G[v].push_back(Edge(u,0,SZ(G[u])-1));
21     }
22     int DFS(int p,int flow){
23         if(p==t) return flow;
24         for(int &i=iter[p];i<SZ(G[p]);i++){
25             Edge &e=G[p][i];
26             if(e.c>0&&d[p]==d[e.v]+1){
27                 int f=DFS(e.v,min(flow,e.c));
28                 if(f){ e.c-=f; G[e.v][e.r].c+=f; return
29 f; }
30             }
31             if(--gap[d[p]]==0) d[s]=tot;
32             else{ d[p]++; iter[p]=0; ++gap[d[p]]; }
33             return 0;
34         }
35     }
36     int flow(){
37         int res=0;
38         for(res=0,gap[0]=tot;d[s]<tot;res+=DFS(s,INF));
39         return res;
40     } // reset: set iter,d,gap to 0
41 } flow;

```

12.4 Bounded Max Flow

```

1 // Author: CRYPTOGRAPHER
2 // Max flow with lower/upper bound on edges
3 // use with ISAP, l,r,a,b must be filled
4 int in[N],out[N],l[M],r[M],a[M],b[M];
5 int solve(int n, int m, int s, int t){
6     flow.init(n+2,n,n+1);
7     for(int i=0;i<m;i++){
8         in[r[i]]+=a[i]; out[l[i]]+=a[i];

```

```

9     flow.addEdge(l[i], r[i], b[i]-a[i]);
10    // flow from l[i] to r[i] must in [a[i], b[i]]
11 }
12 int nd=0;
13 for(int i=0; i <= n; i++){
14     if(in[i]<out[i]){
15         flow.addEdge(i, flow.t, out[i]-in[i]);
16         nd+=out[i]-in[i];
17     }
18     if(out[i]<in[i])
19         flow.addEdge(flow.s, i, in[i]-out[i]);
20 }
21 // original sink to source
22 flow.addEdge(t, s, INF);
23 if(flow.flow()!=nd) return -1; // no solution
24 int ans=flow.G[s].back().c; // source to sink
25 flow.G[s].back().c=flow.G[t].back().c=0;
26 // take out super source and super sink
27 for(size_t i=0; i<flow.G[flow.s].size(); i++){
28     Maxflow::Edge &e=flow.G[flow.s][i];
29     flow.G[flow.s][i].c=0; flow.G[e.v][e.r].c=0;
30 }
31 for(size_t i=0; i<flow.G[flow.t].size(); i++){
32     Maxflow::Edge &e=flow.G[flow.t][i];
33     flow.G[flow.t][i].c=0; flow.G[e.v][e.r].c=0;
34 }
35 flow.addEdge(flow.s, s, INF); flow.addEdge(t, flow.t, INF);
36 flow.reset(); return ans+flow.flow();
37 }

```

12.5 MCMF

```

1 // Author: CRYPTOGRAPHER
2 // Usage:
3 // 1. MCMF.init(n, s, t)
4 // 2. MCMF.add(u, v, cap, cost)
5 // 3. auto [max_flow, min_cost] = MCMF.flow()
6 typedef int Tcost;
7 const int MAXV = 20010;
8 const int INFF = 1000000;
9 const Tcost INFc = 1e9;
10 struct MCMF {
11     struct Edge{
12         int v, cap;
13         Tcost w;
14         int rev;
15         bool fw;
16         Edge(){}
17         Edge(int t2, int t3, Tcost t4, int t5, bool t6)
18             : v(t2), cap(t3), w(t4), rev(t5), fw(t6) {}
19     };
20     int V, s, t;
21     vector<Edge> G[MAXV];
22     void init(int n){
23         V = n;
24         for(int i = 0; i <= V; i++) G[i].clear();
25     }
26     void add(int a, int b, int cap, Tcost w){
27         G[a].push_back(Edge(b, cap, w,
28             (int)G[b].size(), true));
29         G[b].push_back(Edge(a, 0, -w,
30             (int)G[a].size()-1, false));
31     }
32     Tcost d[MAXV];
33     int id[MAXV], mom[MAXV];

```

```

32 bool inqu[MAXV];
33 queue<int> q;
34 pair<int, Tcost> flow(int _s, int _t){
35     s = _s, t = _t;
36     int mxf = 0; Tcost mnc = 0;
37     while(1){
38         fill(d, d+1+V, INFc); // need to use type
39         cast
40         fill(inqu, inqu+1+V, 0);
41         fill(mom, mom+1+V, -1);
42         mom[s] = s;
43         d[s] = 0;
44         q.push(s); inqu[s] = 1;
45         while(q.size()){
46             int u = q.front(); q.pop();
47             inqu[u] = 0;
48             for(int i = 0; i < (int) G[u].size(); i++){
49                 Edge &e = G[u][i];
50                 int v = e.v;
51                 if(e.cap > 0 && d[v] > d[u]+e.w){
52                     d[v] = d[u]+e.w;
53                     mom[v] = u;
54                     id[v] = i;
55                     if(!inqu[v]) q.push(v), inqu[v] = 1;
56                 }
57             }
58             if(mom[t] == -1) break;
59             int df = INFF;
60             for(int u = t; u != s; u = mom[u])
61                 df = min(df, G[mom[u]][id[u]].cap);
62             for(int u = t; u != s; u = mom[u]){
63                 Edge &e = G[mom[u]][id[u]];
64                 e.cap -= df;
65                 G[e.v][e.rev].cap += df;
66             }
67             mxf += df;
68             mnc += df*d[t];
69         }
70         return make_pair(mxf, mnc);
71     }
72 };

```

12.6 Hopcroft-Karp

```

1 // Author: Gino
2 // Function: Max Bipartite Matching in O(V sqrt(E))
3 // Usage:
4 // >>> init(nx, ny, m) -> add(x, y (+nx))
5 // >>> hk.max_matching() := the matching plan
6 // stores in mx, my
7 // >>> hk.min_vertex_cover() := the vertex cover
8 // plan stores in vcover
9 // (!) vertices are 0-based: X = [0, nx), Y = [nx,
10 // nx+ny)
11 struct HopcroftKarp {
12     int n, nx, ny;
13     vector<vector<int>> > G;
14     vector<int> mx, my;
15     void init(int _nx, int _ny) {
16         nx = _nx, ny = _ny;
17         n = nx + ny;
18         G.clear(); G.resize(n);
19     }
20     void add(int x, int y) {
21         G[x].emplace_back(y);

```

```

19     G[y].emplace_back(x);
20 }
21 int max_matching() {
22     vector<int> dis, vis;
23     mx.clear(); mx.resize(n, -1);
24     my.clear(); my.resize(n, -1);
25
26     function<bool(int)> dfs = [&](int x) {
27         vis[x] = true;
28         for (auto& y : G[x]) {
29             int px = my[y];
30             if (px == -1 ||
31                 (dis[px] == dis[x]+1 &&
32                  !vis[px] && dfs(px))) {
33                 mx[x] = y;
34                 my[y] = x;
35                 return true;
36             }
37             return false;
38         };
39
40         while (true) {
41             queue<int> q;
42             dis.clear(); dis.resize(n, -1);
43             for (int x = 0; x < nx; x++){
44                 if (mx[x] == -1) {
45                     dis[x] = 0;
46                     q.push(x);
47                 }
48                 while (!q.empty()) {
49                     int x = q.front(); q.pop();
50                     for (auto& y : G[x]) {
51                         if (my[y] != -1 && dis[my[y]] == -1) {
52                             dis[my[y]] = dis[x] + 1;
53                             q.push(my[y]);
54                         }
55                     }
56                     bool brk = true;
57                     vis.clear(); vis.resize(n, 0);
58                     for (int x = 0; x < nx; x++)
59                         if (mx[x] == -1 && dfs(x))
60                             brk = false;
61                     if (brk) break;
62                 }
63                 int ans = 0;
64                 for (int x = 0; x < nx; x++) if (mx[x] != -1)
65                     ans++;
66                 return ans;
67             }
68             vector<int> vcover;
69             int min_vertex_cover() {
70                 int ans = max_matching();
71                 vcover.clear();
72
73                 vector<int> vis(n, 0);
74                 function<void(int)> dfs = [&](int x) {
75                     vis[x] = true;
76                     for (auto& y : G[x]) {
77                         if (y == mx[x] || my[y] == -1 || vis[y])
78                             continue;
79                         vis[y] = true;
80                         dfs(my[y]);
81                     }
82                 };
83             };

```

```

81         for (int x = 0; x < nx; x++) if (mx[x] == -1)
82             dfs(x);
83         for (int x = 0; x < nx; x++) if (!vis[x])
84             vcover.emplace_back(x);
85         for (int y = nx; y < nx + ny; y++) if (vis[y])
86             vcover.emplace_back(y);
87         return ans;
88     }
89 }
90 }

```

12.7 Cover / Independent Set

- 1 最大邊獨立集 (Ie) 就是最大匹配 (M)
- 2 二分圖上, M 和 Cv 對偶
- 3 對任何圖都有 $|Iv| + |Cv| = |V|$
- 4 對任何圖都有 $|Ie| + |Ce| = |V|$
- 5 二分圖最小帶權點覆蓋 $\Rightarrow$ 建模 (s, u, w[u]) (u, v, INF) (v, t, w[v]) 算最小割

12.8 Kuhn Munkres

```

1 // Author: CRYPTOGRAPHER
2 static const int MXN=2001;// 1-based
3 static const ll INF=0x3f3f3f3f;
4 struct KM{ // max weight, for min negate the weights
5     int n, mx[MXN], my[MXN], pa[MXN]; bool
6     vx[MXN], vy[MXN];
7     ll g[MXN][MXN], lx[MXN], ly[MXN], sy[MXN];
8     void init(int _n){
9         n=_n; for(int i=1; i<=n; i++)
10             fill(g[i], g[i]+n+1, 0);
11     }
12     void addEdge(int x, int y, ll w){ g[x][y]=w; }
13     void augment(int y){
14         for(int x, z; y; y=z)
15             x=pa[y], z=mx[x], my[y]=x, mx[x]=y;
16     }
17     void bfs(int st){
18         for(int i=1; i<=n; ++i) sy[i]=INF, vx[i]=vy[i]=0;
19         queue<int> q; q.push(st);
20         for(;;){
21             while(q.size()){
22                 int x=q.front(); q.pop(); vx[x]=1;
23                 for(int y=1; y<=n; ++y) if(!vy[y]){
24                     ll t=lx[x]+ly[y]-g[x][y];
25                     if(t==0){
26                         pa[y]=x;
27                         if(!my[y]){ augment(y); return; }
28                         vy[y]=1, q.push(my[y]);
29                     }else if(sy[y]>t) pa[y]=x, sy[y]=t;
30                 }
31             }
32             ll cut=INF;
33             for(int y=1; y<=n; ++y)
34                 if(!vy[y] && cut>sy[y]) cut=sy[y];
35             for(int j=1; j<=n; ++j){
36                 if(vx[j]) lx[j]-=cut;
37                 if(vy[j]) ly[j]+=cut;
38                 else sy[j]-=cut;
39             }
40             for(int y=1; y<=n; ++y) if(!vy[y] && sy[y]==0){
41                 if(!my[y]){ augment(y); return; }
42                 vy[y]=1, q.push(my[y]);
43             }
44         }
45     }
46     ll solve(){
47         fill(mx, mx+n+1, 0); fill(my, my+n+1, 0);
48         fill(ly, ly+n+1, 0); fill(lx, lx+n+1, -INF);
49         for(int x=1; x<=n; ++x) for(int y=1; y<=n; ++y)

```

```

45     lx[x]=max(lx[x],g[x][y]);
46     for(int x=1;x<=n;++x) bfs(x);
47     ll ans=0;
48     for(int y=1;y<=n;++y) ans+=g[my[y]][y];
49     return ans;
50 }
51 }graph;

```

13 Combinatorics

13.1 Catalan Number

$$C_0 = 1, C_n = \sum_{i=0}^{n-1} C_i C_{n-1-i}, C_n = C_n^{2n} - C_{n-1}^{2n}$$

0	1	1	2	5
4	14	42	132	429
8	1430	4862	16796	58786
12	208012	742900	2674440	9694845

13.2 Bertrand's Ballot Theorem

- A always $> B$: $C(p+q, p) - 2C(p+q-1, p)$
- A always $\geq B$: $C(p+q, p) \times \frac{p+1-q}{p+1}$

13.3 Burnside's Lemma

Let X be the original set.

Let G be the group of operations acting on X .

Let X^g be the set of x not affected by g .

Let X/G be the set of orbits.

Then the following equation holds:

$$|X/G| = \frac{1}{|G|} \sum_{g \in G} |X^g|$$

14 Special Numbers

14.1 Fibonacci Series

1	1	1	2	3
5	5	8	13	21
9	34	55	89	144
13	233	377	610	987
17	1597	2584	4181	6765
21	10946	17711	28657	46368
25	75025	121393	196418	317811
29	514229	832040	1346269	2178309
33	3524578	5702887	9227465	14930352

$$f(45) \approx 10^9, f(88) \approx 10^{18}$$

14.2 Prime Numbers

- $\pi(n) \equiv$ Number of primes $\leq n \approx \frac{n}{(\ln n)-1}$
- $\pi(100) = 25, \pi(200) = 46$
- $\pi(500) = 95, \pi(1000) = 168$
- $\pi(2000) = 303, \pi(4000) = 550$
- $\pi(10^4) = 1229, \pi(10^5) = 9592$
- $\pi(10^6) = 78498, \pi(10^7) = 664579$